

Detector for Atomic Hydrogen

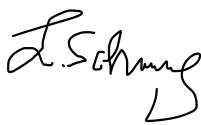
by

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I assure that I have composed this paper independently and no source materials or aids other than those mentioned have been used, and all citations are declared appropriately.

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1. Introduction and Motivation

The purpose of this project was to develop a *detector for atomic hydrogen*, which is suitable not only to detect the presence of atomic hydrogen, but can measure properties of an atom beam, like atom flux or beam profile. Such a device is needed within the Triton-Radius Experiment (T-REX), that aims to test bound state QED and to measure properties of the tritium nucleus. [1]

However, the detector is not restricted to T-REX and can also be used in all other experiments where one uses atomic hydrogen, deuterium or tritium, since it is designed for a general purpose.

The basic idea of T-REX is to create a cold beam of atomic tritium, of which the slower part of the beam will be guided by a magnetic multipole and then slowed down with a cloud of much colder lithium atoms, which are used as a buffer gas in a magneto optical trap (MOT). So that finally it is possible to trap the atoms in a specific area, the MOT, where high-precision laser spectroscopy will be executed with these tritium atoms. In the end T-REX wants to improve the measurement of the triton charge radius by a factor of 300 and gives an opportunity to study forces between the three nucleons of tritium.

Since one wants to use the radioactive element tritium for the final experiment, but does not want to contaminate the setup during tests, atomic hydrogen will be used for the first stages of the experiment. This is possible, because tritium is a hydrogen isotope and thus behaves similarly.[4]

To understand the behaviour of the beam used and to properly verify the performance of the different parts of the experimental setup, as the atom source or the magnetic multipole, it is necessary to analyse the beam properties at different positions within the final experiment. Therefore, a detector is needed, that will be able to measure the beam characteristics, such as the flow and density of the beam, and its shape. Most important the detector should be capable of telling if there is atomic hydrogen, later tritium, present within the beam or not. So the detector will be used in this experiment especially for investigating the setup and the beam to improve the overall performance of the experiment.

The current well understood ways of measuring the former mentioned beam properties are using a quadrupole mass spectrometer (QMS) or the compression tube method. [17] The latter uses a chamber where one measures the pressure with a compression tube and estimates the desired beam properties from the compression tube parameters and the pressure inside the chamber. The biggest disadvantages of these methods are the destruction of the total beam, so it cannot be used for further experiments, the strong requirements on the vacuum when using the QMS [17], and the space

1. Introduction and Motivation

these devices need within the vacuum chamber. Hence, the goal is to overcome these problems and to build a detector that preserves most of the beam, so it can be further used in the experiment. Furthermore, it should be small enough to fit into the vacuum chamber, between the other parts of the experiment.

The general idea for this detector, as Vassiliev et al. proposed in [19], is to use a thin wire (diameter in the μm -regime) to capture the recombination energy of the hydrogen atoms deposited on the wire, during recombination of the atoms on the wire surface to molecular hydrogen. This energy heats the used wire, what eventually leads to an increase of the wire resistance, which one can measure for instance with a Wheatstone bridge.

Therefore, in this thesis the needed theoretical background of the total experiment, the theory of the working principle of the detector, as well as the theory of the electronic parts used will be explained. After dealing with the fundamental theory needed for this thesis, the design of the detector will be presented and explained. Furthermore, the first measurements, done with a laser, will be analysed to show the performance of the detector, and it will be examined, whether it is possible to measure beam profiles. Finally, the results of the analysis will be used to suggest an enhanced version that overcomes problems in the current design.

In sum, this thesis aims to be a reference for future projects, which provides a rather simple and inexpensive detector, that can be used to analyse atomic hydrogen beams, as well as deuterium and tritium.

2. Theoretical Background

2.1. Triton-Radius Experiment (T-REX)

The realm within which this thesis takes place is T-REX. As mentioned before this experiment aims to provide a high precision spectroscopic measurement of atomic tritium, which eventually will result in a 300-times more accurate triton charge radius. With this result it is possible to test predictions from QED. It is also interesting to study tritium, because there is no high-precision spectroscopy done on tritium yet. A better understanding of tritium would help to understand the differences between hydrogen, deuterium and helium, as these atoms were already analysed with the help of high-precision spectroscopy. Additionally, it would allow for precision studies of isospin effects [2]. As mentioned before, H-atoms are used in the first stages of the experiment. One can do this since tritium consists of two neutrons and one proton, thus the main difference between hydrogen and tritium is their mass, which will not affect the performance of the experiment in a critical way. The reason for this is that most of the exploited effects depend on the charge of the atom and only scale with the mass. Therefore, from here on, only hydrogen will be considered in the equations and the text.

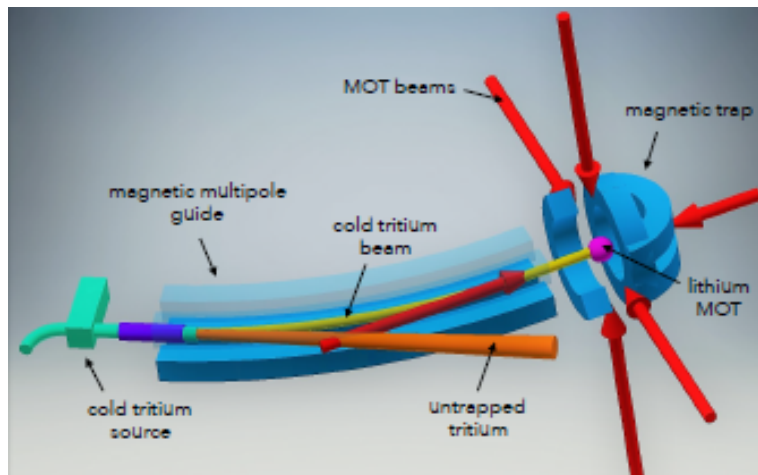
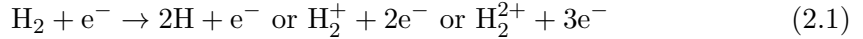


Figure 2.1.: The figure shows a schematic of the planned setup for T-REX. On the left, the atom source is shown, followed by the magnetic multipole to guide the created atom beam. Finally, the atoms are slowed down and trapped in the magnetic trap on the right side. Taken from [2].

2. Theoretical Background

In the beginning of the experiment it is necessary to create an atom beam that can be used for the later stages of the experiment. This beam is created via a microwave dissociator. The H_2 -gas flows, directed by a small glass pipe, into the discharge region, which is wrapped by the microwave cavity. Here free electrons are created by an electrical discharge. The created free electrons e^- and ions H_2^{x+} ($x \in \{1, 2\}$) are then accelerated by the present standing microwave. If they have gained enough energy they will dissociate other molecules in collisions into neutral H-atoms or ions, creating an avalanche of new particles that again go through the same process.



The hereby generated mixture of atoms, molecules and ions will then flow through this glass pipe, that connects the gas source and the microwave dissociator, into a Teflon coated pipe due to a pressure gradient along the tube. The Teflon coated pipe is used to avoid recombination of the H-atoms on its surface [13]. When the beam reaches the vacuum chamber it will be analysed with the hydrogen detector described in this paper, to check the amount of atoms created with the source. Further, a cryostat with liquid Helium is used to cool the beam to $\sim 4.2\text{K}$. Afterwards, the beam will pass a magnetic quadrupole, that is used to guide the slow atoms of the beam to the next part of the setup.

Due to the fact that most of the charged particles will recombine on their way to the quadrupole, the magnetic field of the quadrupole will only interact with the magnetic moment μ of the single atom. The potential energy of an atom in a magnetic field is given by

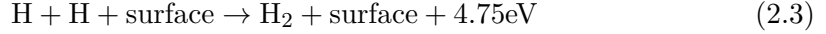
$$E_{\text{mag}} = -\mu_z B = -g_F m_F \mu_B B, \quad (2.2)$$

where B is the magnetic field strength, g_F the Landé-factor of the angular momentum F of the atom, μ_B is the Bohr magneton and m_F is the magnetic quantum number of F , the quantum number of the total angular momentum of the atom. Since m_F is quantised and takes the values $(-1, 0, 1)$ and all other values are constant, there are only two possible non-vanishing values for the potential energy, corresponding to a parallel alignment and an anti-parallel alignment of the magnetic moment relative to the magnetic field. Since the potential energy will be minimized, the atoms with $m_F = 1$ seek the low field and the other the high field. In the middle of the quadrupole there is the lowest field, thus only the low field seekers will follow the quadrupole. Furthermore, only atoms that are slow enough are guided by the quadrupole. Namely, if their kinetic energy is higher than the potential energy barrier due to the magnetic field, they will escape from the quadrupole. After guiding the atoms, it is planned to trap these atoms in a magnetic trap. A cloud of cold lithium atoms confined by a MOT is additionally used to slow down the atoms even more, so it is easier to trap them. The idea is that the atoms lose energy during collisions with the lithium atoms. Finally, laser spectroscopy will be executed with the trapped atoms after removal of the lithium atoms.

2. Theoretical Background

2.2. Hydrogen Recombination on a Metal Surface

To detect atomic hydrogen it is possible to exploit the released energy by the H-atoms when they recombine on a surface to create a signal. When a H-atom hits a surface it may stick to it and stay there. If now a second H-atom meets the bounded one they will recombine to molecular hydrogen and release in total $E_{\text{rec}} = 4.75\text{eV} = 7.60 \cdot 10^{-19}\text{J}$ [19].



This energy can be used for translation, rotation and excitation of the molecule and to heat up the surface. The interesting part here is the heating of the surface, because it can be measured via a Wheatstone bridge.

2.3. Model of the Wire

The above mentioned heating will be measured with a thin wire. The idea is to utilize the fact that the resistance of the wire increases if the wire is heated. This change in resistance will then be measured with a Wheatstone Bridge. To do so, it is inevitable to describe the temperature distribution along the wire, which can be used to calculate the resistance change in dependence of the desired beam properties. In the following, the central ideas for deriving the wire resistance are presented, which can be found entirely in [18]. One can use the one-dimensional differential equation of heat transfer

$$\lambda S \frac{d^2 T}{dx^2} = -F(x, T), \quad (2.4)$$

where $\lambda [\text{W}(\text{mK})^{-1}]$ is the heat conductivity of the wire, S is the cross section of the wire and $F(x, T) [\text{Wm}^{-1}]$ is the one-dimensional density of the heat flow, to describe the one-dimensional temperature distribution along the wire. For further investigation an atom beam with diameter L_B and flux Q hitting the wire with length L_W , diameter d and temperature T_0 perpendicular to its axis is considered, see Fig. 2.2. Further, one has to take into account that a voltage is applied along the wire for measurement purposes.

Putting all of this together, $F(x, T)$ can be written as

$$F(x, T) = F_B(x) + F_{\text{el}}(T) - F_{\text{rad}}(T), \quad (2.5)$$

and the different contributions to F are:

$$F_B = \frac{1}{2} E_{\text{rec}} n = \frac{2Q E_{\text{rec}} d}{\pi L_B} =: C, \quad (2.6)$$

$$F_{\text{el}} = I^2 \cdot (R_0 + R_0 \alpha \cdot [T - T_0]), \quad (2.7)$$

$$F_{\text{rad}} = 4\pi\sigma\epsilon d T_0^3 \cdot (T - T_0). \quad (2.8)$$

Here F_B is the one dimensional density of the recombination heat flow, with n the number of atoms hitting the wire. F_B is assumed to be constant over the illuminated area

2. Theoretical Background

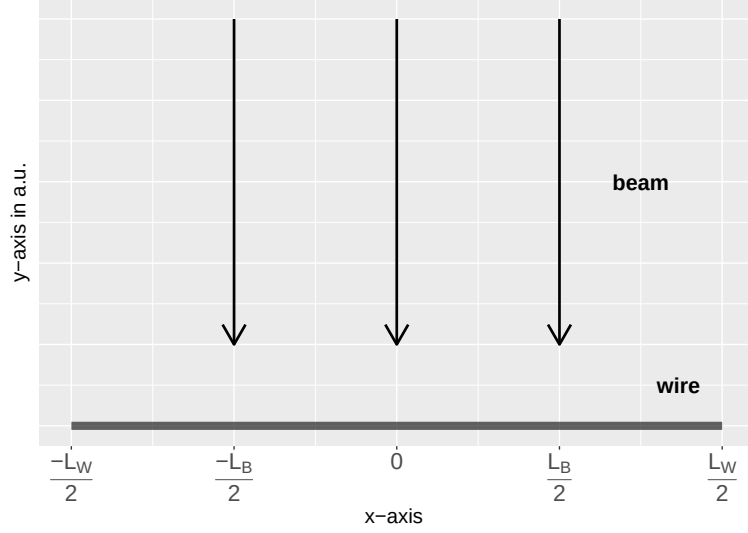


Figure 2.2.: Illustration of the considered beam with diameter L_B and wire with length L_W and diameter d . Surrounding temperature is T_0 .

$(-L_B/2 < x < L_B/2)$ and else zero $(-L_W/2 < x < -L_B/2 \text{ or } L_B/2 < x < L_W/2)$. F_{el} is the heating due to the current I going through the wire, with $R_0[\Omega\text{m}^{-1}]$ being the resistance of the wire per length at temperature T_0 and $\alpha[\text{K}^{-1}]$ the temperature coefficient for the resistance. F_{rad} is the heat loss due to radiation, with $\sigma[\text{Jm}^{-2}\text{s}^{-1}\text{K}^{-4}]$ being the Stefan-Boltzmann constant and ϵ the emissivity of the wire.

Using now fixed boundary condition, which means that the temperature at the wire edges $\pm L_W/2$ is fixed to T_0 , and the symmetry of the setup, one gets a solution by plugging all of this into eq. 2.4. The solution is separated into three parts, of which each describes a different region of the wire, but each of them merge continuously with adjacent solutions, since the temperature distribution along the wire is continuous. In the end the temperature distribution becomes

$$T(x) = T_0 + \frac{\beta_2}{A} + C_1 \cosh(\sqrt{A}x) + C_2 \sinh(\sqrt{A}x) \quad (2.9)$$

$$T(x) = T_0 + \frac{\beta_1}{A} + C_3 \cosh(\sqrt{A}x) \quad (2.10)$$

$$T(x) = T_0 + \frac{\beta_2}{A} + C_1 \cosh(\sqrt{A}x) - C_2 \sinh(\sqrt{A}x). \quad (2.11)$$

2. Theoretical Background

Here the first and last equation describe the distribution in the regions where $F_B = 0$ and the middle one where $F_B = C$. The used constants are

$$\begin{aligned}
A &= \frac{4\pi\sigma\epsilon d T_0^3 - I^2 R_0 \alpha}{\lambda S}, \quad \beta_1 = \frac{C + I^2 R_0}{\lambda S}, \quad \beta_2 = \frac{I^2 R_0}{\lambda S}, \\
C_1 &= \frac{y_0 + (y_0 - y_1) \sinh(x_1) \sinh(x_0)}{\cosh(x_0)}, \\
C_2 &= (y_0 - y_1) \sinh(x_1), \\
C_3 &= \frac{y_0 + (y_0 - y_1)(\sinh(x_1) \sinh(x_0) - \cosh(x_1) \cosh(x_0))}{\cosh(x_0)}, \\
x_0 &= \sqrt{A} \frac{L_W}{2}, \quad x_1 = \sqrt{A} \frac{L_B}{2}, \quad y_0 = -\frac{\beta_2}{A}, \quad y_1 = -\frac{\beta_1}{A}.
\end{aligned} \tag{2.12}$$

With this it is possible to calculate the resistance of the wire by plugging the obtained distributions into the temperature dependency of the resistance, which is

$$R = \int_{-L_W/2}^{L_W/2} R_0 [1 + \alpha(T(x) - T_0)] dx. \tag{2.13}$$

The results are

$$R_B = R_0 L_W + R_0 \alpha \left\{ -y_1 L_B - y_0 (L_W - L_B) + \frac{2(y_0 \sinh(x_0) - (y_0 - y_1) \sinh(x_1))}{\sqrt{A} \cosh(x_0)} \right\} \tag{2.14}$$

when the beam is hitting the wire, and becomes

$$R_{\text{Boff}} = R_0 L_W + R_0 \alpha \left\{ -y_1 L_W + \frac{2y_0 \sinh(x_0)}{\sqrt{A} \cosh(x_0)} \right\} \tag{2.15}$$

when the beam is off.

This change in resistance will be the signal I will measure. Using the Wheatstone bridge, it will be transformed into a voltage, which can then be detected for instance by an oscilloscope.

2.4. Wheatstone Bridge

The Wheatstone bridge circuit can be used to make precise measurements of resistances. This is needed to resolve the shape of the beam, because the changes in the atom flux along the beam profile are presumably small, but need to be recorded for the beam profile. A schematic of the Wheatstone bridge circuit is shown in figure 2.3 below.

A detailed description of the bridge can be found in e.g. [10]. The first important result from this description is, that a completely symmetric bridge, where $R_i = R$ ($i \in \{1, 2, 3, 4\}$), is most sensitive in measuring small changes ΔR in one of the resistances.

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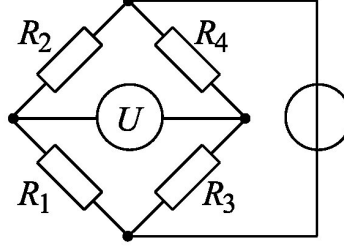


Figure 2.3.: Schematics of the Wheatstone bridge taken from [12]. R_i are the resistor values and U is a voltage.

The second needed result are the equations 2.16 to 2.18 below. In order to measure ΔR one has to balance the bridge in the beginning. This means one has to adjust the bridge such that the signal voltage U becomes zero. In eq. 2.16 one sees that the signal becomes zero if $R_2 R_3$ equals $R_4 R_1$. Commonly, this balancing is done by using potentiometers in one or two of the bridge arms to meet the condition. If one of the resistor values changes after balancing, a voltage U is created across the bridge arms, see fig 2.3. This voltage U is related to R_i and ΔR via the equations

$$U = U_{\text{sup}} \frac{R_2 R_3 - R_4 R_1}{(R_2 + R_4)(R_1 + R_3)}, \quad (2.16)$$

$$U = U_{\text{sup}} \frac{k}{R_2(1+k)} \frac{\Delta R_2}{(1+k) + \frac{\Delta R_2}{R_2}} \quad (2.17)$$

$$\approx U_{\text{sup}} \frac{\Delta R}{4R} \quad (\text{for completely symmetrical case}), \quad (2.18)$$

where U_{sup} is the supply voltage of the bridge and $k = R_3/R_1 = R_4/R_2$. Eq. 2.17 describes the dependency of U on a change of R_2 to $R_2 + \Delta R$, whereas the other resistances stay unchanged.

Due to the fact, that the created signal from the Wheatstone bridge will presumably be small, it is necessary to use some electronic devices to magnify the signal and protect it from noise. Those will be explained in the next section.

2.5. Electronics

The used electronic devices are basic integrated circuits (IC) like differential amplifiers or implemented circuits used for a special purpose. Those are explained shortly in the following.

2.5.1. Operational Amplifier

Operational Amplifiers (Op amps) are used in general to amplify a voltage or a voltage difference compared to reference value, mostly ground. These devices have commonly, as shown in figure 2.4a, two terminals for the input voltages ($V_{\pm}$) and for their power

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supplies ($V_{S\pm}$) and a terminal for the reference value as well as for the output voltage V_{out} . The factor by which the input is amplified is called gain and for most amplifiers it is adjustable. Important characteristics of those ICs are their offset voltage and its drift, the power supply rejection ratio, the gain bandwidth product, the power supply rejection ratio and the noise density.

The offset voltage is the voltage, which needs to be applied to the input terminals so that the output voltage is zero. Its drift describes the temperature dependency of the offset voltage and the power supply rejection ratio is a measure for how much a change in the supply voltage changes the output voltage. Typically, it states how much the offset voltage has to be changed when the supply changes so the output stays the same. The gain bandwidth product is a constant that describes the low-pass behaviour of the amplifier. It is the product of the bandwidth and the gain. The bandwidth is the width of the frequency region with which signals can pass the device unattenuated [15]. The noise density is a measure for the noise pickup of the amplifier. With this it is possible to calculate how much noise will be created by unwanted signals with a given frequency range at the input channel [7]. All of these characteristics can be found listed in the data sheet of a specific device.

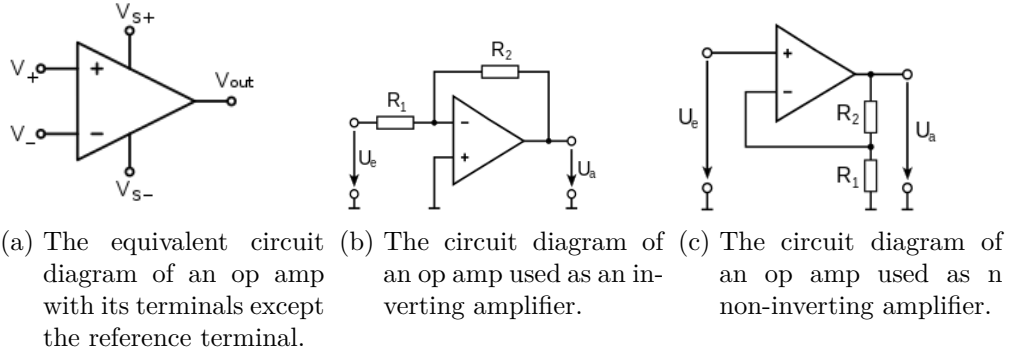


Figure 2.4.: $V_{S\pm}$ are supply voltages, U_e and $V_{\pm}$ are the input voltages, U_a and V_{out} are the output voltages, R_i are resistors. All figures are taken from [3]

In figures 2.4b and 2.4c there are the typical circuit diagrams for an amplifier with feedback shown. As the name implies the non-inverting circuit does not invert the input and the inverting one inverts the polarity of the input. The gain G of such amplifiers is calculated by

$$G = \frac{U_a}{U_e} \quad \text{and becomes}$$

$$G = -\frac{R_2}{R_1} \quad \text{for inverting amplifiers and is} \quad (2.19)$$

$$G = 1 + \frac{R_2}{R_1} \quad \text{for non-inverting amplifiers.} [16] \quad (2.20)$$

2. Theoretical Background

If one wants to amplify a voltage difference, as the signal of the Wheatstone bridge, it is convenient to use an instrumentation amplifier (Ina). This device is an IC as well and has the same characteristics as the afore mentioned op amps, since it consists of different types of op amp circuits. Most of these devices have two extra terminals to adjust the gain individually by placing a resistor between these terminals, and the formula for calculating the gain can be found in the data sheet of the device.

When several amplifiers are used successively the total gain is the product of the individual gains, no matter which type of amplifier is used.

2.5.2. Low-pass Filter

A low-pass filter can be used in a circuit design to attenuate signals with a frequency higher than its so called cut-off frequency f_c and signals with a lower frequency are nearly not affected. Such a filter consists of capacitors, resistors and inductors and uses the dependence of the resistance of capacitors and inductors on the frequency. Its order is estimated by the number of inductors and capacitors and one can implement such filters in an active way by using amplifiers within the circuit or in a passive way without them.

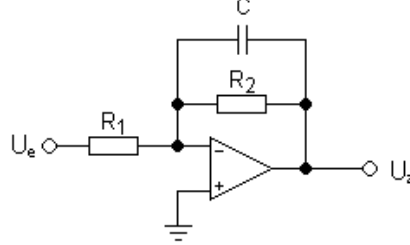


Figure 2.5.: Circuit diagram of a first-order active low-pass filter. U_e is the input voltage, R_i are resistor values, C is a capacitance and U_a is the output voltage. Taken from [11]

As an example a simple first-order active low-pass filter design is depicted in figure 2.5. The gain of this inverting amplifier is given by

$$G = -\frac{R_2}{R_1} \cdot \frac{1}{\sqrt{1 + (\omega C R_2)^2}} \quad [16]. \quad (2.21)$$

In general the cut-off frequency is defined as the frequency at which the output signal amplitude V_{out} is $\frac{1}{\sqrt{2}}$ -times the unattenuated signal V_{in} . In terms of db it is the frequency where the signal loss is equal to 3 dB. The signal change A in dB is calculated by $20 \log \left(\frac{V_{\text{out}}}{V_{\text{in}}} \right)$ [14]. Using this, f_c becomes

$$f_c = \frac{1}{2\pi R_2 C}. \quad (2.22)$$

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So the signal change in dB in dependency on the signal frequency looks like figure 2.6 in a log-plot

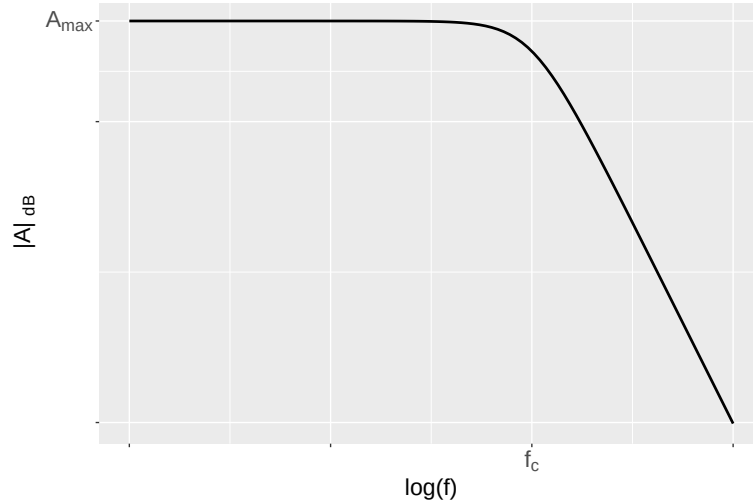


Figure 2.6.: Bode Plot, that shows the signal change against the frequency for a single-pole active low-pass filter, with $A_{\max} = 20 \log(\frac{R_2}{R_1})$ and f_c the cut-off frequency. Please note that the x-axis is in a logarithmic scale.

2.6. Noise

Due to the design of a circuit, imperfection of the used parts or external sources, the observed signal will not be perfect, but instead has some noise on top of it. This noise occurs completely randomly and thus cannot be predicted in its form at any time. Noise is not necessarily positive, it can be negative as well. Hence, it is often the reason for observed fluctuations in circuitry. The quality of a signal can be expressed by the signal-to-noise ratio (SNR), which is given by $\frac{S}{N}$. S is the rms value of the signal and N is the rms value of the Noise without the signal. If noise is plotted separately against time without a signal it will fluctuate around zero in a random pattern. This characteristic allows getting rid of noise by averaging the observed total signal over time without influencing the real signal, if the noise is uncorrelated and the signal can be considered as constant. All the information above were taken from [6].

3. The Experiment

3.1. The Expected Signal

First of all, it is reasonable to calculate an approximate signal to know how sensitive the detector needs to be for the detection of hydrogen.

From equation 2.14 and 2.15, one knows the resistance of the wire used and can, from their difference, compute the resistance change ΔR . Plugging this into equation 2.17 of the Wheatstone bridge gives

$$U_{\text{sig}} = U_{\text{B}} \frac{k}{1+k} \frac{\Delta R}{1+k + \frac{\Delta R}{R_{\text{Boff}}}}. \quad (3.1)$$

Here, U_{B} is the actual voltage drop along the Wheatstone bridge, ΔR the difference between R_{on} and R_{Boff} and $k = \frac{R_{\text{lowerarm}}}{R_{\text{Boff}}} \approx 1$. The following table shows the parameters that are used to calculate an estimation for the signal.

parameter	value
Q	$1 \cdot 10^{14} \text{ s}^{-1}$
L_W	0.035 m
L_B	$5 \cdot 10^{-3} \text{ m}$
d	$25 \cdot 10^{-6} \text{ m}$
α	$4.7 \cdot 10^{-3} \text{ K}^{-1}$
ϵ	0.3
λ	$174 \text{ W} \cdot (\text{mK})^{-1}$
ρ	$0.092 \cdot 10^{-6} \Omega \cdot \text{m}^{-1}$
I	$1.5 \cdot 10^{-3} \text{ A}$

Table 3.1.: Parameters for the signal estimation. α , λ and ϵ are taken from [18], ρ from [9], Q is a guess for the atom flux, which can be expected in the future experiment and all other parameters are taken from the used detector setup cf. tab. A.1.

Inserting these parameters into eq. 3.1, and using that $U_{\text{B}} \approx 2IR_{\text{Boff}}$, gives a signal of $U_{\text{sig}} = 240 \text{ nV}$. The change in the resistance of the wire exposed to the beam is neglected when calculating U_{B} , since it is less than 0.005% of R_{Boff} . Please note that the factor of two in calculating U_{B} is due to the fact that I is only the current through one arm. This estimation serves as a first benchmark for the sensitivity of the detector that should be reached.

But for the sake of simplicity first tests will be done with a laser.

3. The Experiment

3.2. The Laser

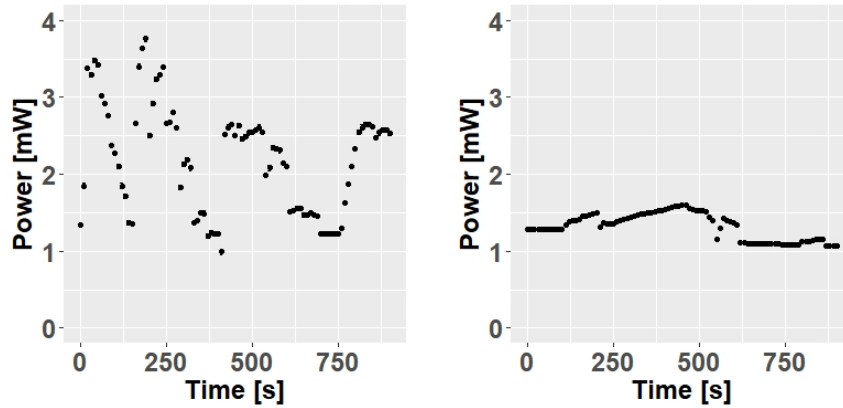
As mentioned above, a laser diode was used to generate a test signal. Therefore, one has to change F_B from eq. 2.6, which was used to estimate the one dimensional density of the heat flow due to the atom beam. Hence, F_B becomes

$$F_B = (1 - \epsilon_L) \frac{Pd}{A_L}, \quad (3.2)$$

where P [W] is the laser power, A_L [m²] the cross section of the laser beam hitting the wire, d [m] the diameter of the illuminated wire and ϵ_L the reflection coefficient of the wire. In addition to the change of C used in the equations 2.12, one has to adjust the beam diameter L_B to the laser.

For this task a simple green laser pointer with a wavelength of $\lambda = (535 \pm 10)$ nm and a power range of 1 mW to 100 mW was used. It was implemented into a voltage source circuit with a switch. With this it is possible to change the laser power within the limitations of the diode, and to turn it of and on without constantly pressing the laser pen button.

Firstly, the stability of the laser power was analysed. To see how the power of the laser behaves over time, the former was measured with a power meter for 15 minutes and every (10 ± 0.5) s the power value was recorded. The results are depicted in figure 3.1 below.



(a) Measurement of the laser power directly after turning it on and adjusted to $P_0 = (1.3 \pm 0.04)$ mW. (b) Measurement of the laser power 1h after turning it on and adjusted to $P_0 = (1.3 \pm 0.04)$ mW.

Figure 3.1.: Power of the laser in mW, measured for 15 min every (10 ± 0.5) s. Please note that error bars are too small to see. The considered uncertainty for the power is the last digit of the power meter.

In fig. 3.1a the fluctuations of the laser power directly after turning it on, and in fig. 3.1b the fluctuations one hour after turning the laser on are shown. Both lasers were

3. The Experiment

directly adjusted to $P_0 = (1.3 \pm 0.04)$ mW after turning the laser on. One can see that the laser runs more stable after running for one hour. Therefore, before each following measurement, the laser was turned on at least 1h before the actual measurement. Furthermore, it was computed from the data of fig 3.1b, that the uncertainty ΔP for the power is at least 0.3 mW for measurements, which last more than a few seconds without monitoring the power. It is obtained by calculating

$$\begin{aligned} \Delta P &= P_0 - P_x = 0.303 \text{ mW} \quad \text{and} \\ \Delta(\Delta P) &= \sqrt{(\Delta P_0)^2 + (\Delta P_x)^2} = 0.05 \text{ mW}, \end{aligned} \quad (3.3)$$

where P_0 is the initial power and $P_x = (1.603 \pm 0.03)$ mW the biggest deviation from P_0 during the measurement.

After this, the radius of the beam is estimated with the knife-edge-method. In this method, the beam is progressively covered with a sharp knife and the power of the beam is measured behind that knife, see fig. 3.2. For the measurement a knife is fixed to a movable holder and the laser beam is placed such, that the beam is orthogonal to the knife edge. The power meter is placed behind the knife. Then the knife is moved in several steps through the laser beam, until the beam is totally covered. During each step the power is recorded.

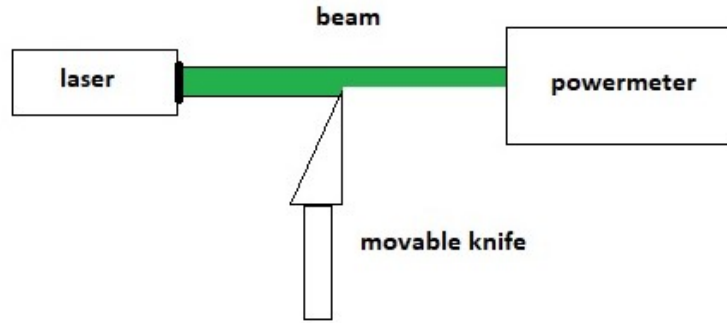


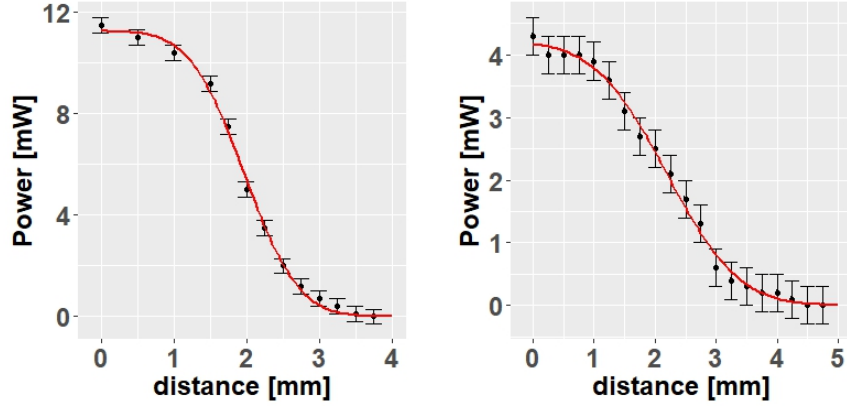
Figure 3.2.: Sketch of the knife-edge method. Either the laser can be moved or the the knife. To detect the laser power a power meter is used.

In fig 3.3 one can see two examples for the obtained data of such a measurement, and a fitted model in red to obtain the $1/e^2$ -radius. The uncertainty for the laser power is taken from the calculation above and the error of the distance is considered to be two division of the micrometer screw, which was used to move the blade. The knife was moved in the same direction, as the laser was moved over the detector. The used model is taken from [8] and is given by

3. The Experiment

$$P(x) = P_{\max} \left\{ 1 - \operatorname{erf} \left[\frac{\sqrt{2}(x - x_0)}{r_{e^{-2}}} \right] \right\}, \quad (3.4)$$

where $P_{\max}$ is the amplitude of the laser power and x_0 is the shift of the beam center and $r_{e^{-2}}$ is the $1/e^2$ -radius of the beam. Therefore, the $1/e^2$ -radius can be extracted from the fit as a fitting parameter, if the beam is considered to have a Gaussian shape in measured direction.



(a) Data from the knife-edge-method with an initial laser power of (11.5 ± 0.3) mW and model fit. Its fit parameter are the one of measurement one in tab. A.2. (b) Data from knife-edge-method with an initial laser power of (4.3 ± 0.3) mW and model fit. Its fit parameter are the one of measurement two in tab. A.2.

Figure 3.3.: Laser power measured in mW for different positions of the knife and in red is the model from eq. 3.4 fitted to the data.

All parameters from the fits are shown in tab. A.2 in the appendix and the corresponding data with fit are shown in the appendix from fig. A.1 to A.6. The radii obtained with this method are depicted in fig. 3.4.

3. The Experiment

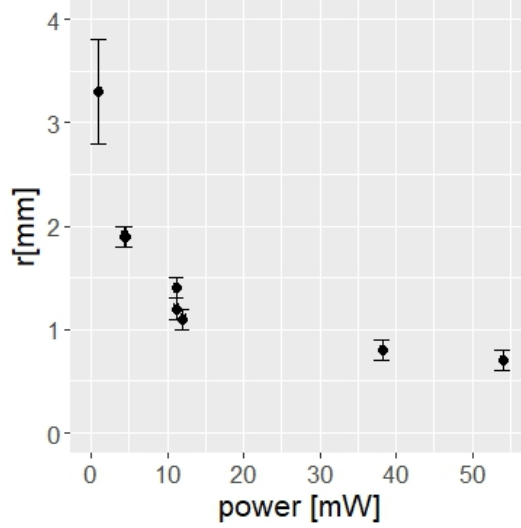


Figure 3.4.: The obtained $1/e^2$ -radii in mm are shown against the used laser power in mW. Please note, that there are two points around 4.5mW.

What one can see in fig. 3.4 is that the $1/e^2$ -radius scales with the used laser power and is getting smaller for higher powers, until it looks like it reaches a plateau at around 0.7 mm. A possible explanation for this could be the modification to the used laser diode, since it is running below its nominal power of 50mW. As a consequence of this, the infrared pump laser, which is used to create the green laser light, has not sufficient energy to fully pump the laser. Thus, some of the infrared light could be emitted from the diode without being used, which eventually could explain the larger radius of the beam, because the infrared light will behave differently than the green light. Also, this could possibly lead to the observed effect that the radius reaches a stable value approaching the nominal power. Additionally, the heat generated by the used voltage source could induce thermal lensing or other effects, causing the beam radius to vary.

Furthermore, a small banana shaped spot besides the main spot of the beam could be observed, when the laser was used with powers below 4 mW. This fact reduces the effective beam diameter L_B hitting the wire, because no light was observed between these spots, and it makes the assumption that the beam has a Gaussian shape questionable. Thus, the obtained data is not fully reliable. Therefore, in future tests a different laser should be used to improve the quality of the obtained data and hence the analysability of these data. However, in order to be able to compare the beam radius obtained with the detector these results are used as references for the beam radius in the following.

Finally, the results obtained above are used to compare the one dimensional heat density of the laser to the one of the considered atom beam, to see if the laser is a suitable replacement. Using the data given in tab. 3.1 for the estimation of F_B , in

3. The Experiment

the case an atom beam is used, results in a value of $2.8 \cdot 10^{-5} \frac{\text{W}}{\text{m}}$. Since this value is just a rough estimation there is no uncertainty computed. If F_B is calculated for the case the laser is used with a power of 1.1 mW it becomes $(12.1 \pm 3.1) \cdot 10^{-5} \frac{\text{W}}{\text{m}}$. The used laser radius is 3.3 mm and the reflection coefficient used is 0.85, since the wire is plated with gold [5]. This comparison shows, that the laser has a more than four times higher one dimensional heat density than the considered atom beam. However, the laser will still be used to see, if the concept of the detector is working, and since the order of magnitude of the F_B 's are similar, the results obtained with the laser can serve as a first benchmark for the performance of the detector.

3.3. The Detector

3.3.1. The Prototype

3.3.1.1. Design of the Prototype

First, a simple prototype was build. A picture of the circuit can be seen in fig. A.12 and A.11 in the appendix. Below one can see the schematics of the circuit of the prototype.

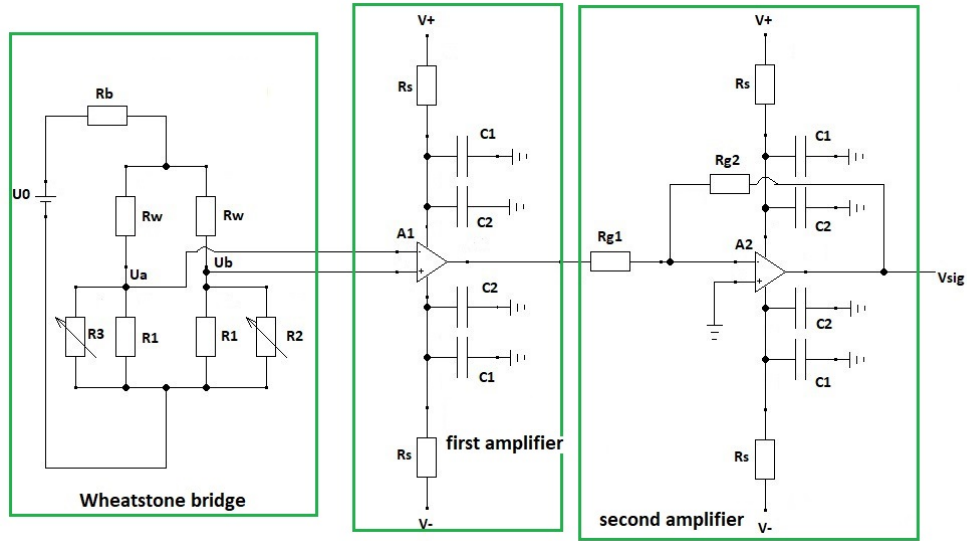


Figure 3.5.: Circuit Design of the prototype. Parts and specifications are listed in tab. A.1. The different sections in the circuit are named below them.

The most left part of the schematics above is the Wheatstone bridge. This bridge is supplied with the voltage $U_0 = 12\text{V}$ and to avoid big currents going through the bridge the resistor R_b is placed in front of it. This is done to reduce the effect of Ohmic heating in the used wires. The R_w 's are the wires from Luma Metall [9] with a diameter $d = (25 \pm 0.25)\mu\text{m}$ and a length L_w of $(3.5 \pm 0.1)\text{cm}$. From that the resistance

3. The Experiment

of the wire was calculated and is $R_w = (6.56 \pm 0.19)\Omega$. Both of the upper bridge arms are wires so that the bridge becomes more symmetric and thus is easier to balance. The lower arms of the bridge consist of simple metal foil resistors called R_1 , parallel to potentiometers (R_3 & R_2), so it is possible to balance the bridge by adjusting the potentiometers. Next to the bridge there is an instrumental amplifier (A_1), whose input terminals are connected to the bridge points U_a and U_b to amplify the voltage difference between them. Its gain is fixed to $G_1 = 1064 \pm 21$ by placing a resistor with $(47 \pm 0.94)\Omega$ resistance between two specific terminals. The resistor is not shown in the schematics and the formula for the gain can be found in the data sheet referenced in section A.4. Now follows another amplification step, where an operational amplifier (A_2) is used as an inverting amplifier. The gain can be derived using eq. 2.19 and becomes $G_2 = 51 \pm 1$, when applying $R_{g1} = (47 \pm 0.94)\Omega$ and $R_{g2} = (2.4 \pm 0.024)\text{k}\Omega$. Thus, the total gain becomes $G_{\text{tot}} = 54332 \pm 1629$. The equations for the uncertainties of the gains can be found in the appendix in eqn. A.1. The negative voltage V_- is -12V and the positive voltage V_+ is 12V . All voltage inputs are stabilised by using the capacitors C_1 and C_2 and the resistors R_s to protect the amplifiers from spikes and gaps in the voltage supply. Finally, the whole circuit was placed into a metal box to protect it from surrounding noise, since the metal box is a Faraday cage. The box has a small opening above the wire, to allow the beam to hit the wire. Further BNC connectors were fixed to the box for the signals and inputs. One connector was used for the signal after the first amplification step and another for the signal after the second stage of amplification. For the power supplies two BNC connectors were used and each of the two bridge points U_a and U_b were connected to BNC as well. The latter was done to enable observing the unamplified bridge signal, while balancing it. A picture of the whole detector is shown in fig. A.14.

3.3.1.2. Measurement Procedure

To create a signal the laser was fixed onto a movable stage and the detector was placed beneath the laser, such that the laser beam could hit the wire and could be moved out of the box. The laser was moved with a micrometer screw, which has a precision of $\pm 10\mu\text{m}$. Before each measurement the bridge was balanced. This was done by observing the voltage U_{ab} between the bridge points U_a and U_b with a multimeter, and adjusting the potentiometers such that $U_{ab} < \pm 1\mu\text{V}$. The upper boundary of $1\mu\text{V}$ was chosen because the multimeter has a precision of exactly this value. Finally, the signal after the amplification steps was recorded with an oscilloscope. The experimental setup can be seen in the sketch below.

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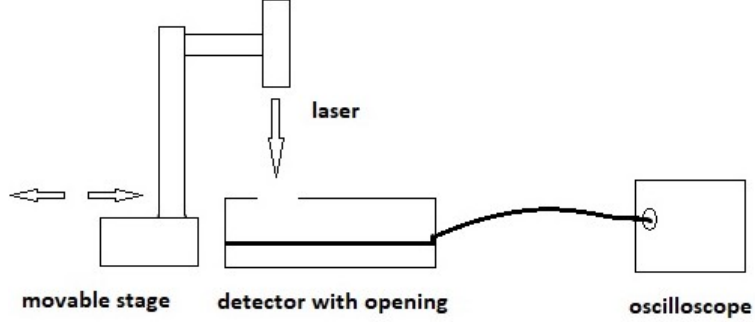


Figure 3.6.: Sketch of the experimental setup. Inside the box is the circuit. The wire on the circuit, which is exposed to the beam, is showing into the paper plane. The laser can be moved to the left and right, and is pointing down.

3.3.1.3. Detector-Performance

First, the raw signal without a laser beam, pre-processing with the oscilloscope and its internal low-pass filter was recorded over 2 s, see fig. 3.7a. Then, the laser beam with a power of 2.5 ± 0.3 mW was placed on the wire to create a signal, which was recorded for 2 s as well, see fig. 3.7b. Every $500 \mu\text{s}$ a data point was obtained, so in total $n_m = 4000$ data points were obtained.

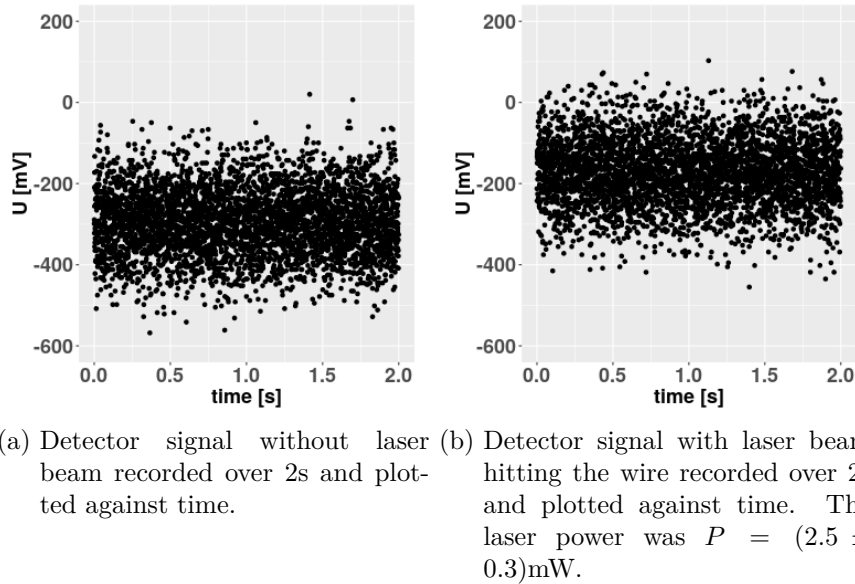


Figure 3.7.: Measured detector signal with different experimental settings. The error is considered to be one division of the oscilloscope scale for U , $\Delta U = 1\text{mV}$. $\Delta t = 1\text{ms}$, it was considered to be more accurate than one division, since the oscilloscope has a higher time precision than $40 \text{ ms} (\equiv \text{one div.})$.

3. The Experiment

From these data the rms value (U_{rms}), the mean (U_{mean}) and its standard deviation (σ) were calculated and are shown in tab. 3.2 below.

data	U_{rms} [mV]	U_{mean} [mV]	σ [mV]
fig. 3.7a	306	-295	81
fig. 3.7b	190	-173	81

Table 3.2.: From the data of fig. 3.7 calculated magnitudes. U_{rms} is the rms value of U , U_{mean} is the mean of it and σ the standard deviation. The standard deviation is considered as a measure for the uncertainty of the former mentioned values.

Since one would expect a mean of 0V if there is no beam on the wire, one can compute from fig. 3.7a that the signal of the detector has an offset of $U_{\text{offset}} = U_{\text{mean}} = (-295 \pm 81)\text{mV}$ at the output channel of the detector, which corresponds to a signal of $(5 \pm 2)\mu\text{V}$ before amplification, see eq. A.3. This can be explained by the offset voltage of the used amplifiers, which can be found in their data sheets. Another reason could be, that the bridge was not perfectly balanced resulting in an offset as well. In the worst case, the offset created by the amplifiers could be around 2.7V in the current setup (A.4). Therefore, for the next design it would be reasonable to implement a technique for canceling the offset of the first amplifier to avoid saturation in the second amplifier, because the offset of the first amplifier will be magnified by the combined gain and the second offset only by its gain.

Further, this observed offset explains why the rms value of the data with the laser signal is lower than the one without it. The laser signal increases the observed voltage, resulting in a smaller absolute value, since the offset shifts the ground level into a negative voltage area. So to compute the actual noisiness of the signal, the obtained data has to be corrected and then the SNR can be calculated from that. First, the offset, mentioned above, was removed from the data of fig. 3.7a and 3.7b by subtracting the mean of fig. 3.7a from the data. Then, a new rms value was computed using the new data and with these the SNR was calculated as defined in section 2.6, and the error was obtained by Gaussian uncertainty propagation. The result is $\text{SNR} = (1.8 \pm 1.0)$. This ratio and its high error ($> 50\%$) show that the data are rather noisy and that the noise makes up more than half the signal.

In addition to that, the obtained signal should be compared to the theoretical value. The theoretical value can be obtained in the same manner as the expected signal from the hydrogen beam in section 3.1. For this task, a $1/e^2$ -radius for a laser power of $(2.5 \pm 0.3)\text{mW}$ was obtained via linear interpolation of the first two data points shown in fig. 3.4. The results for the $1/e^2$ -radius is $r_{e^{-2}} = (2.8 \pm 0.7)\text{mm}$, which will result in a theoretical signal of $(116 \pm 4)\text{mV}$. The measured signal is considered to be the difference between the two mean values shown in tab. 3.2, which is $(122 \pm 115)\text{mV}$. The obtained uncertainty is due to the high σ . Nevertheless, the measured signal matches the theoretical one within the uncertainty.

One can reduce the noisiness of the data by averaging the gathered data over time. To

3. The Experiment

show the effect of different averaging times τ_{average} on the signal quality, the obtained n_m data points from above were divided into subsets of length τ_{average} . Each subset is then averaged over time to create one data point. From this new data set, which contains $\frac{n_m}{\tau_{\text{average}}}$ data points, the mean and the standard deviation were calculated. For example, $\tau_{\text{average}} = 500 \mu\text{s}$ would reproduce the initial data. The result of this approach is depicted in fig. 3.8. The red data represent the data without a laser signal, and the black with laser. The shown error bars are the corresponding sigma to the processed data.

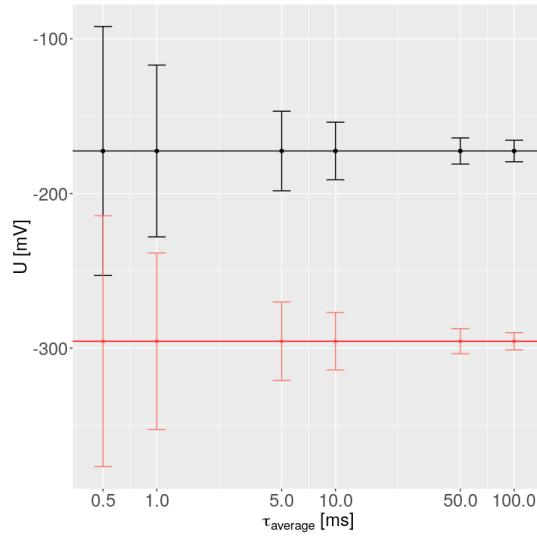


Figure 3.8.: τ_{average} is the bin size used to create one data point, by averaging all points inside the bin. The error bars are the corresponding σ . The red points resemble the results without a laser signal and the black with one. Please note, that the x-axis has a logarithmic scale.

The obtained results from this approach are shown additionally in tab. 3.3.

τ_{average} [ms]	0.5	1.0	5.0	10.0	50.0	100.0
mean ₀ [mV]	-295	-295	-295	-295	-295	-295
mean ₁ [mV]	-173	-173	-173	-173	-173	-173
σ_0 [mV]	81	57	25	19	8	6
σ_1 [mV]	81	56	26	19	8	7

Table 3.3.: The effect of different averaging times τ_{average} on the standard deviation σ . σ_0 and mean₀ show the results for data without the laser signal, in fig. 3.8 red, and σ_1 and mean₁ the results for the data with a laser signal, in fig. 3.8 black.

3. The Experiment

Looking at tab. 3.3 shows, that averaging over different times does not change the mean value of the signal, but it enhances the quality of the signal, which is expressed by the smaller σ due to averaging. Theoretically, it would be possible to reduce σ to arbitrary small values by averaging long enough, but since one is interested in measuring beam profiles the actual change in the signal has to be resolved. During the beam profile measurements, the micrometer screw will be moved with a rotation frequency of 0.1 Hz or less, which corresponds to linear velocity of the laser beam of $v_L = 0.5 \text{ mm} \cdot 0.1 \text{ Hz} = 0.05 \text{ mm}$. Therefore, less than $\frac{v_L}{r_e^{-2}} < 1\%$ of the laser beam will be moved per second over an area. Hence, an averaging time of 1 s or 100 ms will be used in the following.

Further, a sensitivity of the detector can be computed from the data of tab. 3.3. As a criterion for the differentiation between background fluctuations and an actual signal, a $3\text{-}\sigma$ interval is considered. The mean σ for an averaging time of 100 ms is 6.5 mV and thus $3\sigma = 19.5 \text{ mV}$, which corresponds to an unamplified signal of $(360 \pm 11) \text{ nV}$, which is 120 nV higher than the proposed sensitivity in section 3.1. This value can be seen as a first estimation for the sensitivity of the detector. In order to enhance the sensitivity the average time can be increased, but this could lead to problems during measurements if the beam signal changes too fast or the beam is moved too fast. So on the one hand it would be reasonable to increase the gain of the detector in a future design, and on the other hand to reduce the noise present in the setup. A possible way of reducing the noise would be implementing low-pass filters into the circuit design. To conclude from the former, the internal low-pass filter of the oscilloscope is used in the following to improve the signal quality. On top of that, pre-processing is used to generate the data points. This means that one data point represents the mean value of a total time-sweep (1 s or 100 ms), and its uncertainty is the standard deviation of the mean. This can be done, as mentioned before, without worsening the signal, because the signal is a DC-voltage, which does not change fast, and the noise is considered to be random and uncorrelated noise, see section 2.5.2 and 2.6.

In figure 3.9 two measurements are shown, both over a duration of 1000s. During the measurement resembled by fig. 3.9a, the cover of the detector box was removed, and during the other one the box was closed again. One can see that putting the circuit into the box improves the signal quality insofar, as the fluctuations on a short time scale vanish. It can be explained by the fact that the box shields the circuit from outer influences like electro-magnetic radiation or fast thermal fluctuations. Due to this result, all measurements in the following were done with the box covered, and for future designs it is recommended to also place it in a Faraday cage.

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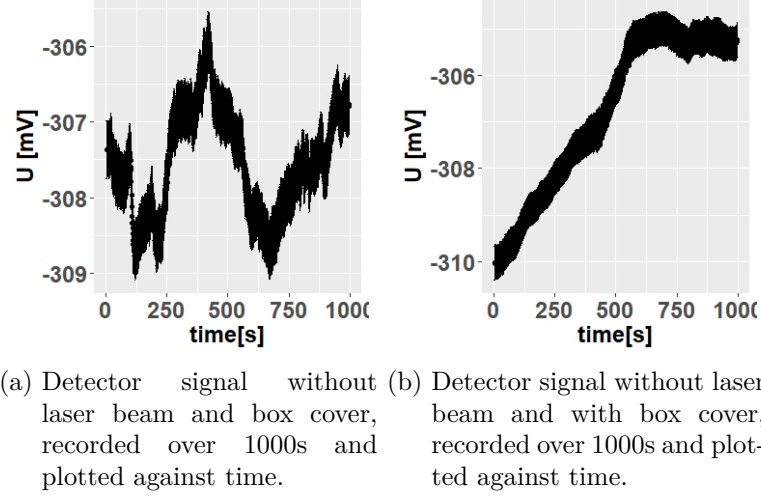


Figure 3.9.: Measured detector signal with different experimental settings. The measuring method is mentioned above in this section.

Further, the signals after the first amplification step and the second amplification step are compared to each other to see if the calculated gain of the second amp is in agreement with the experiment. This is done by measuring both signals simultaneously, then taking the ratio of both, and plotting this ratio against time. The result is shown in fig. 3.10 and a close up of the data can be found in fig. A.7 in the appendix. One would expect that the result should be constant at $G_2 = 51$, but since noise will be picked up in the cables between the detector and the oscilloscope, and there are also instabilities in the gain, the actual data will fluctuate around G_2 . From this data, the median of the ratio and its error was calculated to compensate the fluctuations. The median was chosen in this case, because it is more insensitive against extreme values than the mean and thus will not be affected by the few very high values in the plot. The value of the median is (52 ± 2) , what is in good agreement with the calculated gain $G_2 = (51 \pm 1)$. For comparison, the mean of the ratio is (54 ± 2) , which is still in agreement with G_2 . However, for the first amplifier it was not possible to check its gain, because there was no reference value for the signal before the amplification step.

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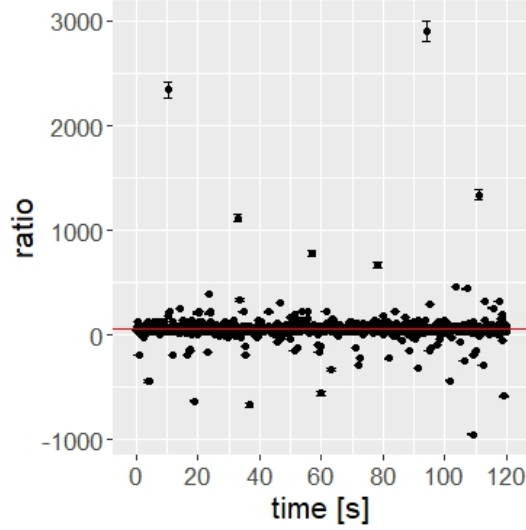


Figure 3.10.: Ratio of the data from the second and first amplifier obtained during the same measurement. The error of the ratio was calculated by Gaussian uncertainty propagation. In red a straight line at 51.

In addition to the afore mentioned measurements, the long-term behaviour of the bridge was observed. To do so, a sweep of 50s was averaged to generate the mean, which is used as one data point, and as the uncertainty the standard deviation of this mean was chosen. The result is depicted in fig. 3.11.

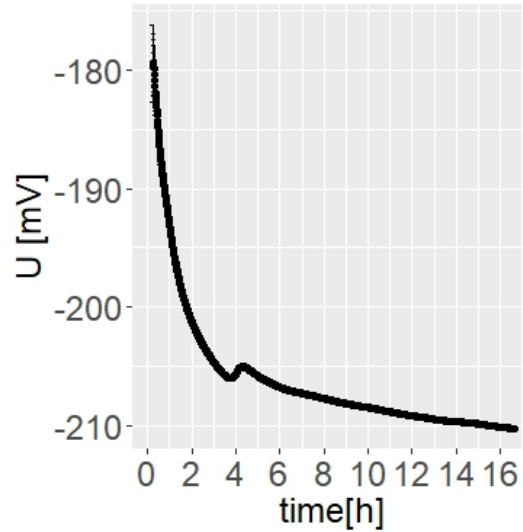


Figure 3.11.: Long-term behaviour of the detector, measured for 16h. One data point is the average over 50s and the uncertainty is its standard deviation.

3. The Experiment

The data were obtained without the laser hitting the wire, but the laser was on during the whole measurement and located next to the detector. At around four hours, the current regulator of the laser circuit was touched to check how warm it was. Due to the additional heat of the hand next to the detector, a signal was created resulting in the kink at around four hours. Beside that, the signal seems to be stable and is not fluctuating on a short time scale. During the total measurement, the signal shrinks by about 30mV. This corresponds to an unamplified signal of $(560 \pm 17)\text{nV}$. A possible reason for this behaviour is the period of the measurement. It was started at 16pm and finished at 8am. Since the laboratory is located in the basement, which is not heated during the night, the room temperature will drop, causing the offset voltage of the amplifiers to drift. A possible temperature drift of 1°C to 3°C would be sufficient enough to cause this drift, because the offset of the first amplifier has a temperature drift of $\pm \frac{200\text{nV}}{^\circ\text{C}}$ up to $\pm \frac{500\text{nV}}{^\circ\text{C}}$. The second amplifier was not taken into account, since its offset is amplified only by $G_2 = (51 \pm 1)$, causing a drift in the μV -regime in the considered case, which is too small to be resolved with the oscilloscope. The temperature dependence of the gain is also not considered, because its maximum drift is $\pm 10 \frac{\text{ppm}}{^\circ\text{C}}$, which is within the uncertainty of the gain.

On top of this, imperfections in the Wheatstone bridge, such as differences in the used resistors or wires due to e.g. manufacturing errors, could cause the parts to behave not identically, which eventually unbalances the bridge. As a result, a signal will be created, which also could explain the observed drift.

3.3.1.4. Powerscaling

To analyse how the detector signal behaves when the power of the laser is changed, the signal created by different laser powers was measured. An example for the obtained data can be seen in fig. 3.12. Each peak in this data set represents a measurement for a different laser power. The red line is the mean value of the signal before the actual measurements were started, and the green lines are the values, which were used to extract the peak height. The lower green values resemble the mean value of data sampled 1 min before the laser was placed on the wire, and green lines in the peaks are the average of five data points around the highest value in the peak. This was done to reduce the influence of fluctuations.

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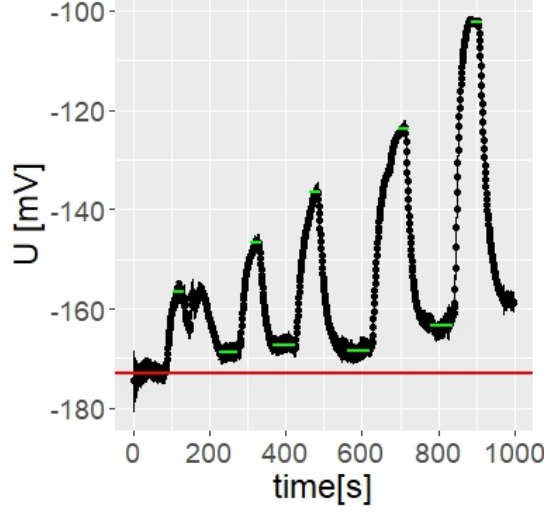


Figure 3.12.: Measured signal created by different laser powers. Each peak represents a measurement with a different laser power. In red the mean of the initial signal, (-173 ± 0.6) mV. In green the values, which were used to extract the peak height.

The peak height is considered to be the difference between the peak value and the lower green value. The uncertainties of the green values are considered to be their standard deviation and the uncertainty for the actual peak height was obtained by Gaussian uncertainty propagation. The results for all ten measured peaks are listed in tab. 3.4. The peak number is also the order in which the data was obtained.

peak	P [mW]	U_{in} [mV]	U_{max} [mV]	ΔU [mV]	U_{cor} [mV]
1	(3.2 ± 0.3)	(-173.0 ± 0.6)	(-156.7 ± 0.1)	(16.3 ± 0.6)	(16.3 ± 0.6)
2	(4.2 ± 0.3)	(-168.8 ± 0.3)	(-146.7 ± 0.1)	(22.1 ± 0.3)	(26.3 ± 0.7)
3	(6.4 ± 0.3)	(-167.3 ± 0.3)	(-136.5 ± 0.2)	(30.8 ± 0.4)	(36.5 ± 0.7)
4	(8.1 ± 0.4)	(-168.5 ± 0.5)	(-123.8 ± 0.5)	(44.7 ± 0.7)	(50.2 ± 0.8)
5	(15.6 ± 0.4)	(-163.4 ± 1.0)	(-102.2 ± 0.6)	(61.2 ± 1.2)	(70.8 ± 1.3)
6	(21.8 ± 0.4)	(-162.9 ± 1.4)	(-100.4 ± 0.1)	(62.9 ± 1.4)	(73.0 ± 1.5)
7	(37.7 ± 0.4)	(-156.5 ± 0.9)	(-98.7 ± 0.1)	(57.8 ± 0.9)	(74.8 ± 1.1)
8	(48.7 ± 0.4)	(-139.5 ± 0.7)	(-98.7 ± 0.1)	(40.8 ± 0.7)	(74.3 ± 0.9)
9	(17.4 ± 0.4)	(-126.6 ± 0.9)	(-99.7 ± 0.7)	(26.9 ± 1.1)	(73.3 ± 1.3)
10	(13.2 ± 0.4)	(-126.3 ± 0.6)	(-99.4 ± 0.3)	(26.9 ± 0.7)	(73.6 ± 0.9)

Table 3.4.: P is the used laser power, U_{in} is the initial value (the lower green value, cf. fig. 3.12), U_{max} is the maximum value from each peak (the upper green one, c.f. fig. 3.12), ΔU is the difference between U_{max} and U_{in} and U_{cor} is ΔU plus the difference between U_{in} and -173.0 mV.

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In fig. 3.13 the data from tab. 3.4 above were used to plot the peak height against the laser power. The black data points are ΔU from tab. 3.4 and the red data points are U_{cor} . ΔU was calculated as described before, using the green marked values in fig. 3.12. U_{cor} is the peak height in relation to the red line in fig. 3.12. A reason for doing so is, that the substrate on which the wire is located is also heated by the laser. Therefore, this substrate functions as a heat bath, which increases the initial temperature of the wire for each measurement. Hence, one can argue that ΔU is not the total signal and one needs to add the difference between the initial value before all measurements (-173.0 ± 0.6) mV and the initial signal before each measurement. Further, the data obtained from peak six and higher are measured more than 20 min after the first value, especially the last measurements. So it could be possible that some drift in the bridge happened, which makes these data points questionable.

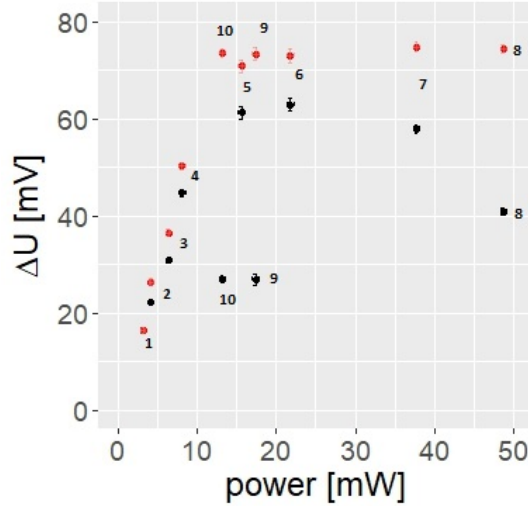


Figure 3.13.: The measured signal in mV plotted against the used laser power in mW. In black ΔU and in red U_{cor} from tab. 3.4. The numbers annotate the peak and the order of the measurements.

One can see in fig. 3.13 that for data points with a number smaller than six the black and red data points behave similarly. For laser powers below 15 mW they follow a linear trend and then it looks like they begin to saturate. After measurement six the red and black data behave differently. The red data saturate at around 70 mV and the black values drop. One reason could be the mentioned possibility of a drift, so that the data points indexed higher than six are not comparable to the first six. Another reason could be that the measurements with higher laser power (six, seven and eight) heated the substrate and thus caused the before mentioned offset. This could explain why the last three measurements have such a low signal and why it is justified to use the red data points only. To see, which explanation fits the observation more, it would be reasonable to repeat this measurement. But for time reasons it was not possible to do this.

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However, the important result is that the detector behaves linearly for small powers. This suggests that the detector will behave similarly for signals smaller than the ones from the laser. This implies that one can build an enhanced version with an increased sensitivity, which will behave similar to the prototype.

3.3.1.5. Beam Profile with the Detector

In order to measure the beam profile of the laser, the total beam was moved over the wire while the signal was recorded. The laser was moved with the micrometer screw in a uniform way. Further, the number N_R of rotations of the screw was counted and the time t_m of the measurement was recorded, so it was possible to compute from this a rotation frequency $\nu_R = \frac{N_R}{t_m}$ for the screw, which will be used to transform the time scale of the measurement into a position information. An example of the data of such a profile scan is depicted in 3.14. The laser was adjusted to $P = (1.7 \pm 0.3)$ mW, the measurement duration was $t_m = (300 \pm 1)$ s and (15 ± 1) rotations were done during this time. Thus $\nu_R = (0.05 \pm 0.003)\text{s}^{-1}$, where the uncertainty was calculated with Gaussian uncertainty propagation.

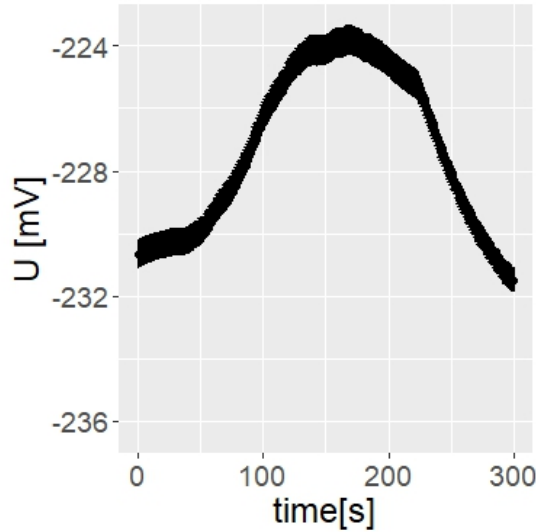


Figure 3.14.: Laser beam profile obtained with the detector. The laser was adjusted to $P = (1.7 \pm 0.3)$ mW, the measurement duration was $t_m = (300 \pm 1)$ s and (15 ± 1) rotations were done during this time.

To obtain the $1/e^2$ -radius of the beam from this measurement, a model was fitted to the data. The model describes a Gaussian beam and in addition to that a linear background. The fluctuations are given, as shown in section 3.3.1.3, in a first approximation by linear drifts. Further, the substrate beneath the wire will be heated by the laser as well and thus the frame temperature of the wire increases, what is also taken into account to be in first order approximation linear. Both effects are consid-

3. The Experiment

ered as the reasons why the signal is not reaching the same magnitude as before the measurement and therefore causing the linear background. The fitted model is

$$U(t) = mt + U_{01} + U_{\max} \exp \left[-\frac{(t - t_0)^2}{w} \right], \quad (3.5)$$

where t is the time, $(mt + U_{01})$ describes the linear background, with m the slope and U_{01} the y-axis interception of it. The last term of eq. 3.5 describes the peak shape, where $U_{\max}$ is a measure for the amplitude of the peak, t_0 is the position of the peak on the x-axis and w is a measure for the width of the peak. If one wants to compute the $1/e^2$ -radius of the beam from this model, only the latter part, which describes the beam, has to be considered in the calculation. The result is

$$r_{e^{-2}} = \sqrt{2w}. \quad (3.6)$$

If one uses the measured data for the computation of the $1/e^2$ -radius, it will describe the radius in the time domain. Hence, it is necessary to transform it to the space domain, which eventually leads to the results

$$r_{e^{-2}} = \sqrt{2w} \nu_R z \quad \text{and} \quad (3.7)$$

$$\Delta(r_{e^{-2}}) = \sqrt{\left(\sqrt{2w} \nu_R \Delta z\right)^2 + \left(\sqrt{2w} \Delta \nu_R z\right)^2 + \left(\frac{\Delta w \nu_R z}{\sqrt{2w}}\right)^2}. \quad (3.8)$$

Z describes the gear ratio of the micrometer screw and is given by (0.5 ± 0.01) mm per rotation, w and its uncertainty are obtained from the fit and ν_R and its uncertainty are calculated as mentioned before. In fig. 3.15 the example from fig. 3.14 is depicted together with the fitted model in red.

3. The Experiment

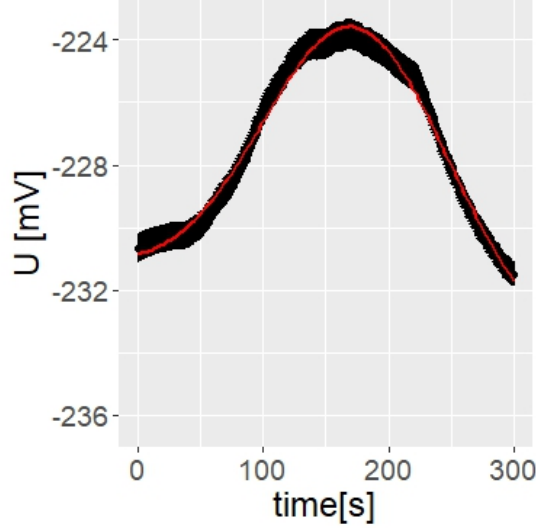


Figure 3.15.: Laser beam profile obtained with the detector. The laser was adjusted to $P = (1.7 \pm 0.3)$ mW, the measurement duration was $t_m = (300 \pm 1)$ s and (15 ± 1) rotations were done during this time. One data point is the mean of one second. In red the fit of the model from eq.3.5.

The fit and measurement parameters are depicted in tab. 3.5 below.

$m[\text{mVs}^{-1}]$	$U_0[\text{mV}]$	$U_{\text{max}}[\text{mV}]$	$t_0[\text{s}]$	$w[\text{s}^2]$	N_R	$t_m[\text{s}]$
(-8.3 ± 0.5)	(-231.5 ± 0.1)	(9.3 ± 0.1)	(173.8 ± 0.7)	(11040 ± 311)	(15 ± 1)	(300 ± 1)

Table 3.5.: Obtained fit parameters (m , U_0 , U_{max} , t_0 , w) from the fit shown in fig. 3.15 and (N_R , t_m) from the measurement. Uncertainties are obtained from the fit and from the measurement.

Plugging w and its uncertainty and the measurement parameters (N_R , t_m) into eq. 3.7 and 3.8, one gets $r_{e-2} = (3.7 \pm 0.2)$ mm. In addition to this measurement the results of two further measurements are shown in fig. A.8 and A.9, and the the corresponding fit parameters in tab. A.3. All results from these measurements are summarised in tab. 3.6 below.

Measurement	fig. 3.15	fig. A.8	fig. A.9	knife-edge method
$r_{e-2}[\text{mm}]$	(3.7 ± 0.2)	(3.7 ± 0.3)	(4.0 ± 0.4)	(3.5 ± 0.5)
$P [\text{mW}]$	(1.7 ± 0.3)	(1.5 ± 0.3)	(1.7 ± 0.3)	(1.1 ± 0.3)

Table 3.6.: The obtained results for the $1/e^2$ -radii of the laser using the detector and the knife-edge method.

If one compares these results to each other, they show the reproducibility within the uncertainties. Furthermore, the obtained radii are comparable with the ones obtained

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from the knife-edge method, see tab. 3.6. Therefore, it can be stated that it is possible to measure the beam profile with the detector.

3.3.2. First Iteration of the Prototype

The results from the characterisation of the prototype were used to design an enhanced version, the first iteration. The mentioned parts are chosen for a printed circuit design to make the detector more compact and easier to build. In the following, these improvements are shown and explained. Since the various sections of the new design are shown individually, a total circuit diagram of the first iteration can be found in fig. A.10 in the appendix, as well as a table with the chosen parts (tab. A.4).

To begin with, the circuit of the Wheatstone bridge is shown in fig. 3.16.

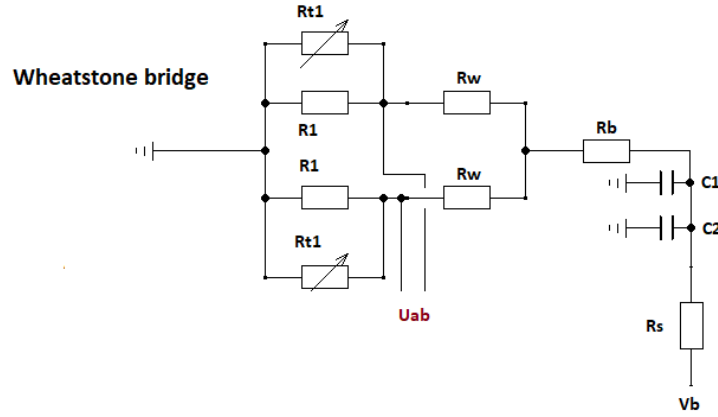


Figure 3.16.: The Wheatstone bridge circuit design of the first iteration. R are Resistors, C capacitors and V and U Voltages.

First of all, it is planned to stabilize the input voltage V_b of the bridge in the same way as for the power supplies of the amplifiers in the prototype design. It is done to reduce fluctuations in the bridge. The capacitors used have the values $C_1 = 0.1\mu\text{F}$ and $C_2 = 10\mu\text{F}$ and $R_s = 100\Omega$. Further, R_b is planned to be exchangeable, so it is possible to test different values. R_w represents the wire, which will be also from Luma Metall. The wire is going to have a diameter of $d = 20\mu\text{m}$, which is smaller than the one in the first design, as this will increase the signal [18]. It will have a length of 3 cm. To make the bridge more symmetric one array resistor will be used for both R_1 . An array resistor is one part containing several resistors with same resistance, manufactured out of one piece, see its data sheet referenced in section A.4. The potentiometers R_{t1} will be both trimmers with the same maximum resistance. U_{ab} is the signal voltage and connection to the first amplifier should be as short as possible and both lines should travel the same distance to avoid asymmetries and

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signal loss. Finally, it is planned that the wire hit by the beam will be outside the detector box and the substrate beneath it will be removed to avoid destroying the beam or heating of the substrate.

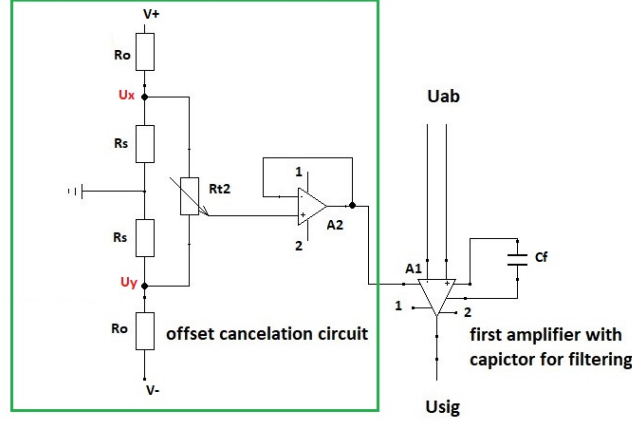


Figure 3.17.: The circuit design of the first amplifier of the first iteration. R are Resistors, C capacitors, A amps and V and U Voltages. Note, that 1 and 2 are the power supply pins of the amplifiers, but the circuit is the same as in the prototype and thus omitted.

In fig. 3.17, the first amplification step is shown. The signal U_{ab} will be amplified by the instrumental amplifier A_1 . Its gain is internally fixed to 2000. Furthermore, it has the option to be used as a low-pass filter if a capacitor is placed between two specific pins. In this case, C_f is used to accomplish this. The capacitance of C_f is considered to be $0.56 \mu\text{F}$, which yields a cut-off frequency of 47.4 Hz. For the equation see the data sheet of A_1 . Due to this cut-off frequency, every signal change faster than $\frac{1}{47.4\text{Hz}} = 20\text{ms}$ will be attenuated. However, it will not affect the signal since it is a DC-signal changing slower, and on top of this, the process of averaging makes it impossible to resolve changes faster than the averaging time. The last improvement to the first amplification step is the offset cancellation circuit. It will be fixed to the reference pin off A_1 , so it is possible to change the reference value within a fixed range to cancel the offset voltage of A_1 . An instruction for this technique can be found in the data sheet of A_1 . The idea is to use a voltage divider between the voltages V_+ and V_- . The reference value is then fixed between U_x and U_y and can be varied by the potentiometer R_{t2} . The range is fixed by the resistors R_o , R_{t2} and R_s and the voltages V_+ and V_- and is given by

$$U_x = -U_y = V_+ \frac{R_s R_{t2}}{2R_s R_o + (R_s + R_o)R_{t2}}. \quad (3.9)$$

$U_x = -U_y$, because the circuit is symmetric and $V_+ = -V_-$, in the considered design. Next, the second and third amplifier stage will be discussed, cf fig. 3.18. It is consid-

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ered to implement a third amplification step, which will be optional and looks similar to the second one except for its gain, which will be smaller to avoid saturation of the signal.

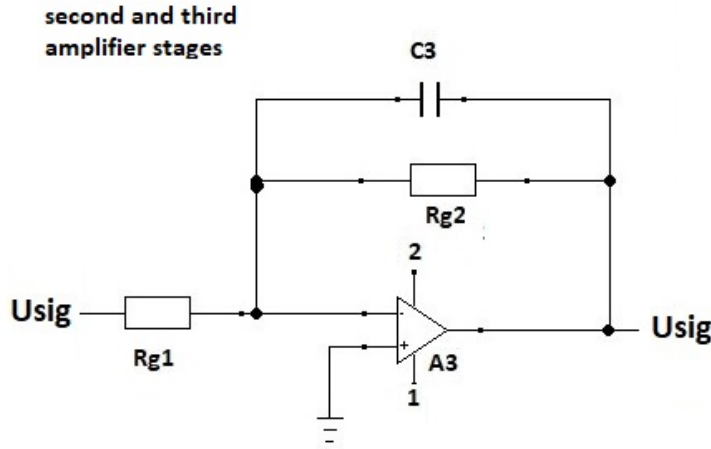


Figure 3.18.: The circuit design of the second and third amplifier of the first iteration. R are Resistors, C capacitors, A amps and V and U Voltages. Note, that 1 and 2 are the power supply pins of the amplifiers, but the circuit is the same as in the prototype and thus omitted.

A_3 is used as an inverting low-pass filter in the second and third stage. The values of the parts are chosen to be $R_{g1} = 10\Omega$, $R_{g2} = 10k\Omega$ and $C_3 = 0.53\mu$ for the second stage. Therefore, its gain is given by 1000 and the cut-off frequency by 28.4 Hz, see eq. 2.21 and 2.22. For the third stage only, the value of R_{g1} is changed to 1 k Ω . Thus, the gain of the third stage is 10 and the cut-off frequency remains unchanged. The use of the third stage will be optional, depending on which signal will be better, and whether the third stage will saturate. In general, the gain was increased so it is more likely that the hydrogen beam will create a detectable signal. Remember that the hydrogen beam will create a smaller signal than the laser.

All of the above mentioned changes to the detector design are due to the obtained results from the prototype. In order to see whether these considered changes will truly enhance the detector performance, the suggested design should be built and tested in a similar way as the prototype was tested. Encouragingly, the obtained results and the theory suggest that the improvements will enhance the detector performance.

4. Summary and Outlook

First of all, the examination of the used laser suggests that in future experiments a different laser setup should be used, which produces a better shaped beam and can be adjusted in its power in a more reproducible way, so it allows a better understanding of the beam and makes the obtained data more reliable. Nevertheless, a laser can be used in principle to simulate a hydrogen beam, since the fundamental principle of the detector stays unchanged and both beam types heat the wire. Furthermore, the used laser generates signals higher than those expected from the hydrogen beam. Thus, a laser setup should be considered, which generates signals with the same order of magnitude as the hydrogen beam will do. However, since the used laser produces signals, albeit about four times higher than the expected signal, still in a comparable regime, the results can be seen as a first proof of concept, which can be used to build an enhanced version of the presented prototype.

Further, the analysis of the used detector shows that signals in the nV-regime can be detected and that it is possible to reach the desired sensitivity of 240 nV by using the idea proposed in the beginning. It was also possible to verify that the used theory can predict the magnitude of the signal in an appropriate way that matches the observation. Therefore, the theory can be used to make estimations, which can be taken into account during the detector design. On top of that, it could be shown that, with the current detector design, it is feasible to measure beam profiles of the laser, and from these estimate beam radii. These are in agreement with the results obtained with the knife-edge method. Hence, it can be concluded that using a Wheatstone bridge and several amplification steps is a suitable way of designing the detector. But in order to be able to analyse a hydrogen beam, it is necessary that the current design of the prototype will be improved. The results suggest that the gain should be increased, since the signal from the hydrogen beam is smaller than the one from the laser. Because of the increased gain, saturation in the used amplifiers should be considered as a possible problem. Furthermore, the substrate beneath the wire, which will be heated by the beam, should be removed, so undesired effects as background heating or reflection of atoms is prevented. Additionally, one should increase the stability of the Wheatstone bridge. This can be achieved by making the Wheatstone bridge as symmetric as possible. It is reasonable to use low-pass filter to suppress noise. In addition to these mentioned results, an enhanced version of the prototype was designed, which takes the results of this thesis into account in order to overcome the before mentioned problems.

For the future, it is planned to build the enhanced version of the detector and to test its performance in a similar way it was done in this thesis with the prototype. Some

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further experiments should be also done with the new version, like testing the total gain with a function generator to verify the calculations for the gain and to see if the phase of the signal changes. Additionally, it should be tested if the design will be vacuum compatible, since it will be used inside a vacuum chamber. Beside testing the detector with a laser, there will be first measurements with a hydrogen beam to fully prove that the detector can accomplish the task of analysing a hydrogen beam. After the first hydrogen beam measurements it would be necessary to design a moving mechanism, which enables the measurement of a hydrogen beam profile, within the used vacuum chamber.

A. Appendix

A.1. Tables

name	part	specification
U_0	power supply	+12V
V_+	power supply	+12V
V_-	power supply	-12V
R_b	metal foil resistor (MR)	$(3.9 \pm 0.039) \text{ k}\Omega$
R_W	wire wire from luma metals	$d = 25\mu\text{m}, l = 3.5\text{cm}, [9]$
R_1	MR	$(20 \pm 0.2)\Omega$
R_2	potentiometer	$(20 \pm 0.2)\Omega$
R_3	potentiometer	$(50 \pm 2.5)\Omega$
R_s	MR	$(100 \pm 1)\Omega$
R_{g1}	MR	$(47 \pm 0.94)\Omega$
R_{g2}	MR	$(2.4 \pm 0.024)\text{k}\Omega$
A_1	INA 128 P	data sheet
A_2	OPA 37 GP	data sheet
C_1	tantalum capacitor	$10\mu\text{f}$
C_2	foil capacitor	$0.1\mu\text{f}$

Table A.1.: Used parts and their characteristics in the prototype. Referenced data sheets can be found in section A.4.

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measurement	$P_{in}[mW]$	$P_{max}[mW]$	$x_0[mm]$	$r_{e-2}[mm]$
1	11.5 ± 0.3	11.3 ± 0.2	2.0 ± 0.1	1.2 ± 0.1
2	4.3 ± 0.3	4.2 ± 0.1	2.2 ± 0.1	1.9 ± 0.1
3	1.1 ± 0.3	1.8 ± 0.4	0.8 ± 0.4	3.3 ± 0.5
4	11.2 ± 0.3	11.5 ± 0.3	1.2 ± 0.1	1.4 ± 0.1
5	4.5 ± 0.3	5.7 ± 0.3	0.8 ± 0.1	1.9 ± 0.1
6	12 ± 0.3	11.7 ± 0.1	1.9 ± 0.1	1.14 ± 0.1
7	54 ± 0.3	52.4 ± 0.5	1.3 ± 0.1	0.7 ± 0.1
8	38.2 ± 0.3	36.7 ± 0.4	1.7 ± 0.1	0.8 ± 0.1

Table A.2.: Obtained fit parameters (P_{max} , x_0 , r_{e-2}) from the knife-edge method measurements of the beam radius and their errors. P_{in} in $[mW]$ is the initial power of the laser to which it was adjusted. The data can be found in 3.3 and from fig. A.1 to A.6.

data	fig. 3.15	fig. A.8	fig. A.9
$m[mVs^{-1}]$	$(-8.3 \pm 0.5) \cdot 10^{-3}$	$(-1.25 \pm 0.07) \cdot 10^{-2}$	$(-1.24 \pm 0.09) \cdot 10^{-2}$
$U_0[mV]$	(-231.5 ± 0.1)	(-228.7 ± 0.1)	(-232.9 ± 0.1)
$U_{max} [mV]$	(9.3 ± 0.1)	(6.6 ± 0.2)	(2.7 ± 0.1)
$t_0[s]$	(173.8 ± 0.7)	(147.8 ± 0.9)	(78.2 ± 0.9)
$w[s^2]$	(11040 ± 311)	(7858 ± 358)	(3181 ± 231)
N_R	(15 ± 1)	(11 ± 1)	(13 ± 1)
$t_m [s]$	(300 ± 1)	(187 ± 2)	(130 ± 1)

Table A.3.: Obtained fit parameters (m , U_0 , U_{max} , t_0 , w) from the fits shown in fig. 3.15, A.8 and A.9 and (N_R , t_m) from the measurement. Uncertainties are obtained from the fit and from the measurement.

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name	part	specification
V_b	power supply	+12V
V_+	power supply	+15V
V_-	power supply	-15V
R_b	TFR, SMD	variable, to test
R_W	wire wire from luma metals	$d = 20\mu m, l = 3cm$, [9]
R_1	CRA06P08322R0FTA	$(22 \pm 0.22)\Omega$
R_{t1}	3223W-1-200E Bourns	$(20 \pm 2)\Omega$
R_{t2}	3224W-1-103E Bourns	$(10 \pm 1)k\Omega$
R_s	CRCW1206100RFKEAC	$(100 \pm 1)\Omega$
R_o	ERJ-2RHD3902X	$(39 \pm 0.19)k\Omega$
R_{g1}	ERJ2RKD10R0X	$(10 \pm 0.05)\Omega$
R_{g2}	CRCW120610K0DHEAP	$(10 \pm 0.05)k\Omega$
R_{g3}	RC1206JR-7W1KL	$(1000 \pm 10)\Omega$
R_{g4}	CRCW120610K0DHEAP	$(10 \pm 0.05)k\Omega$
A_1	AD 842 BRZ	data sheet
A_2	OP 1177 ARMZ-R7	data sheet
A_3	OPA 2188 AID	data sheet
A_4	OPA 2188 AID	data sheet
C_1	SMDTC03100KA00MQ00	$(0.1 \pm 0.02)\mu f$
C_2	TAJA106K016RNJV	$(10 \pm 1)\mu f$
C_3	ECW-U1564KCV	$(0.56 \pm 0.056)\mu f$

Table A.4.: Used parts and their characteristics in the design of the first iteration.
TFR is short for thick film resistor and SMD for surface mounted device.

A.2. Figures

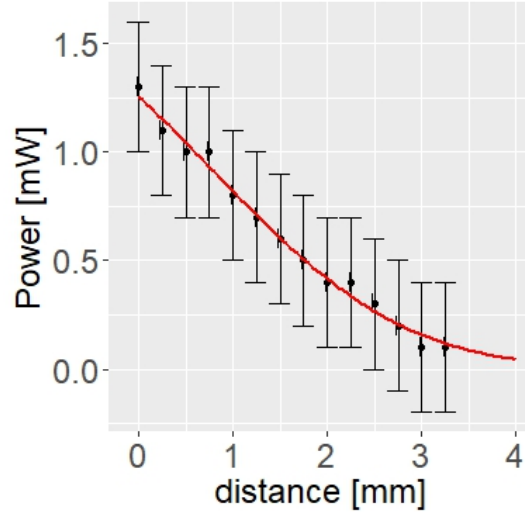


Figure A.1.: Data from the knife-edge-method with an initial laser power of $(1.1 \pm 0.3)\text{mW}$ and model fit. Its fit parameter are the on of measurement 3 in tab. A.2.

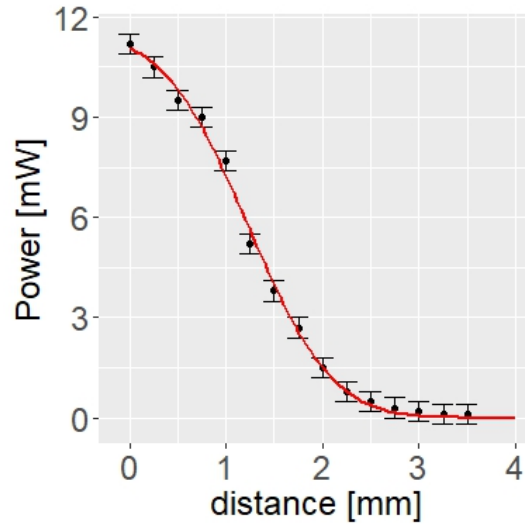


Figure A.2.: Data from the knife-edge-method with an initial laser power of $(11.2 \pm 0.3)\text{mW}$ and model fit. Its fit parameter are the on of measurement 4 in tab. A.2.

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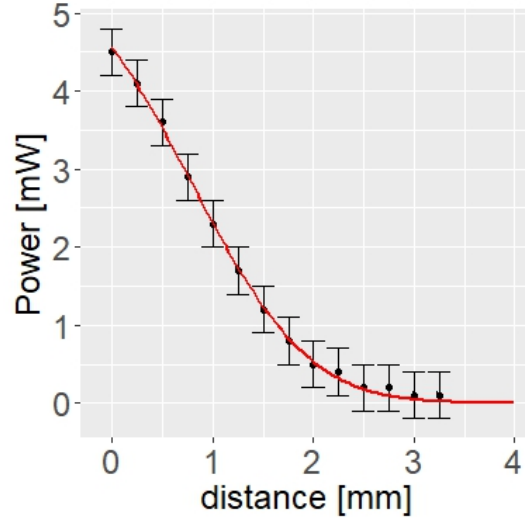


Figure A.3.: Data from the knife-edge-method with an initial laser power of $(4.5 \pm 0.3)\text{mW}$ and model fit. Its fit parameter are the on of measurement 5 in tab. A.2.

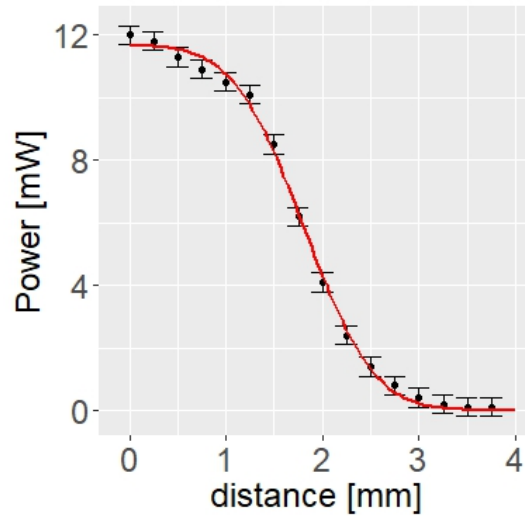


Figure A.4.: Data from the knife-edge-method with an initial laser power of $(12 \pm 0.3)\text{mW}$ and model fit. Its fit parameter are the on of measurement 6 in tab. A.2.

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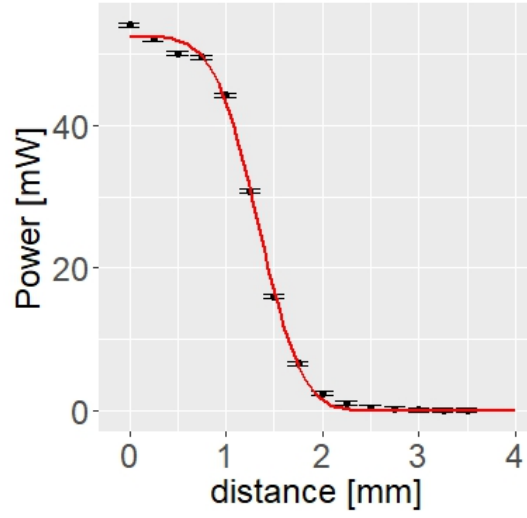


Figure A.5.: Data from the knife-edge-method with an initial laser power of $(54 \pm 0.3)\text{mW}$ and model fit. Its fit parameter are the on of measurement 7 in tab. A.2.

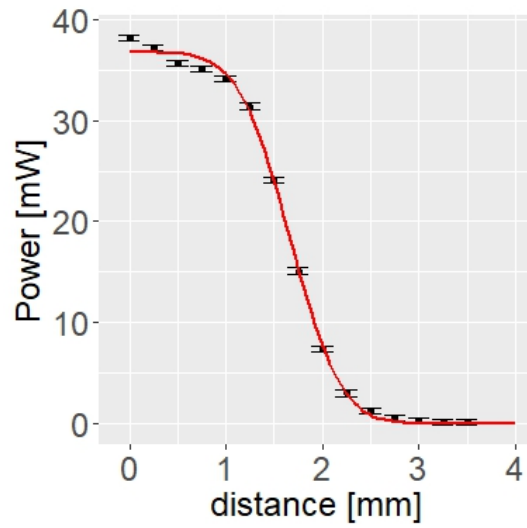


Figure A.6.: Data from the knife-edge-method with an initial laser power of $(38.2 \pm 0.3)\text{mW}$ and model fit. Its fit parameter are the on of measurement 8 in tab. A.2.

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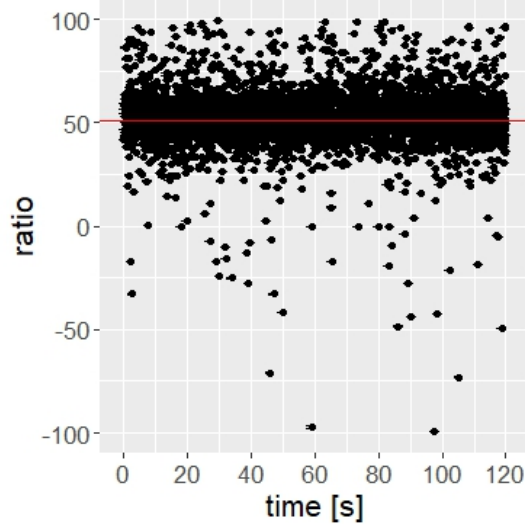


Figure A.7.: Ratio of the data from the second and first amplifier obtained during the same measurement. The error of the ratio was calculated by Gaussian error propagation. In red a straight line at 51. Please note, that this is close up of the data and a full scale plot is shown in fig. 3.10.

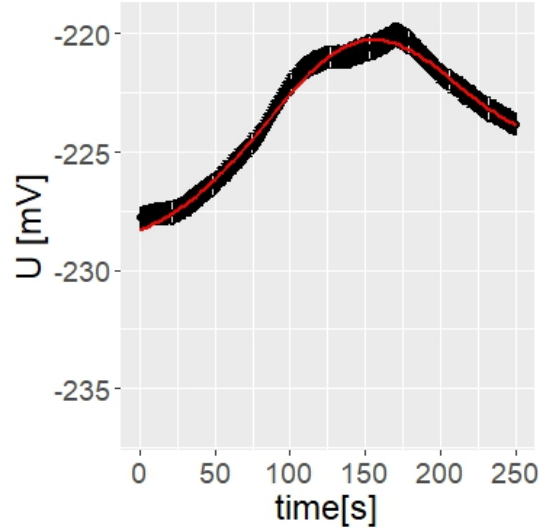


Figure A.8.: Laser beam profile obtained with the detector. The laser was adjusted to $P=(1.5 \pm 0.3)$ mW, the measurement duration was $t_m = (187 \pm 2)$ s and (11 ± 1) rotations were done during this time. One data point is the mean of one second. In red the fit of the model from eq.3.5.

A. Appendix

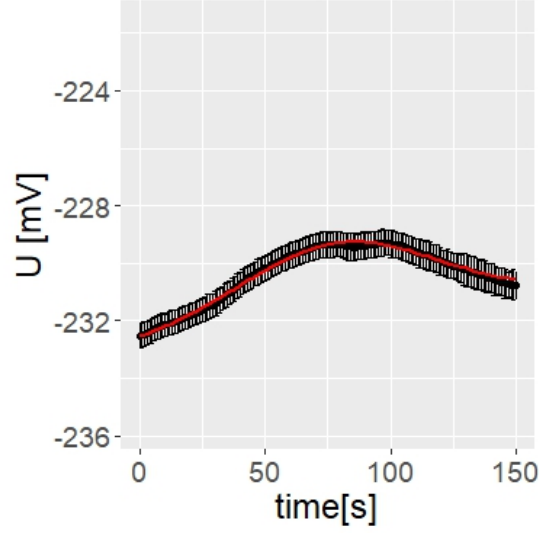


Figure A.9.: Laser beam profile obtained with the detector. The laser was adjusted to $P=(1.7 \pm 0.3)$ mW, the measurement duration was $t_m = (130 \pm 1)$ s and (13 ± 1) rotations were done during this time. One data point is the mean of one second. In red the fit of the model from eq. 3.5.

A. Appendix

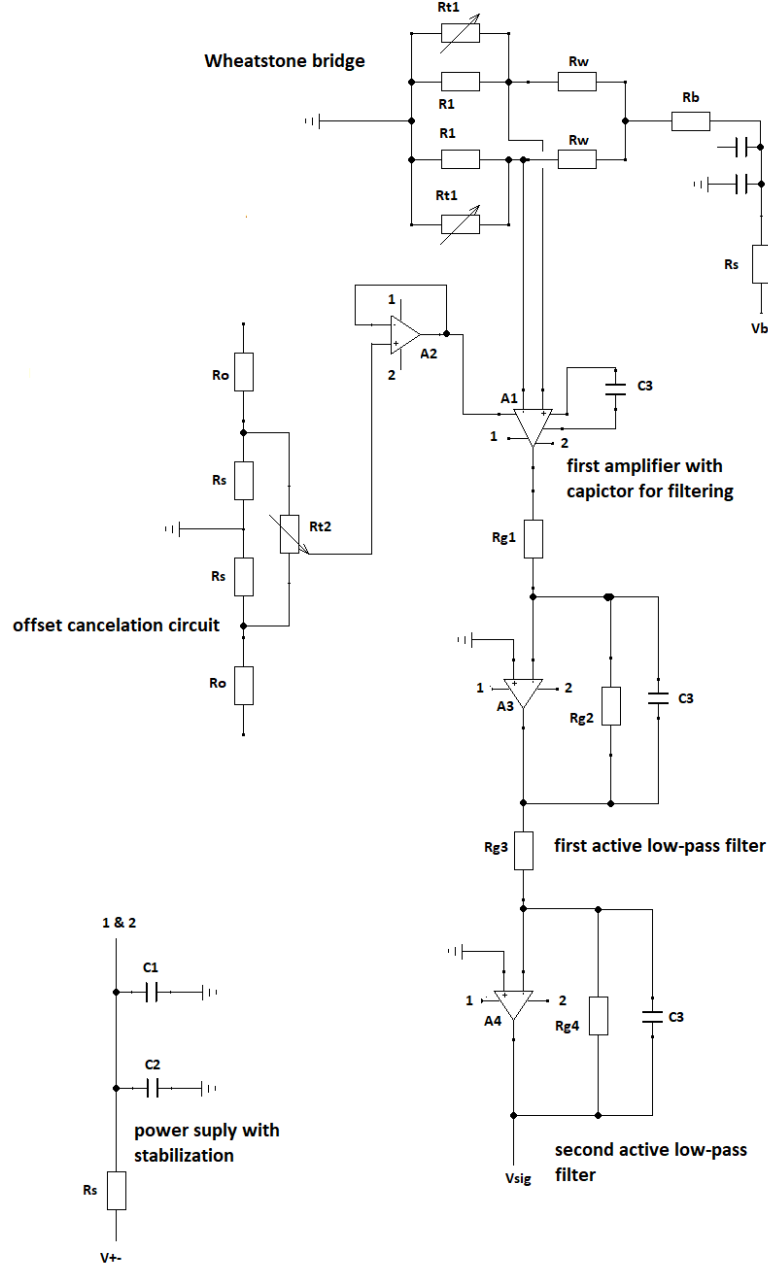


Figure A.10.: The total circuit design of the first iteration of the prototype. R are Resistors, C capacitors, A amps and V Voltages. Please note, for the sake of clarity the circuit for the power supply of the amps is individually drawn in the left corner. A list of the parts and their specifications can be found in tab. A.4.

A. Appendix

A.3. Pictures

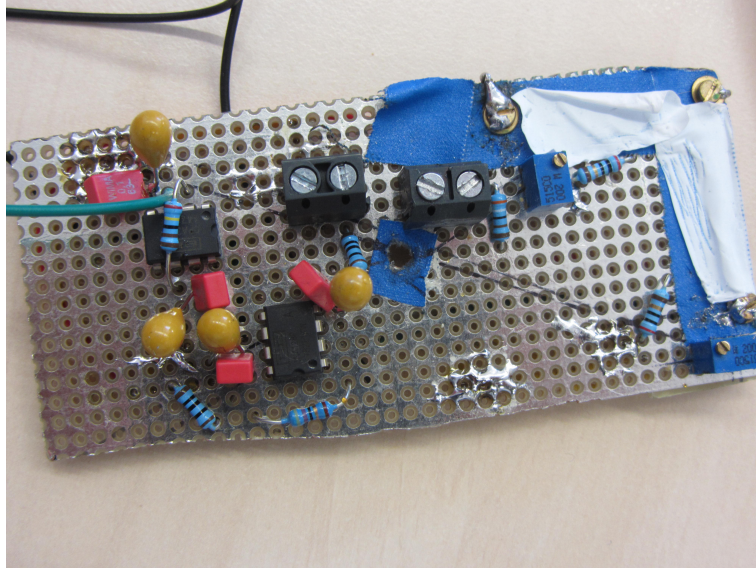


Figure A.11.: Top view on to the build prototype

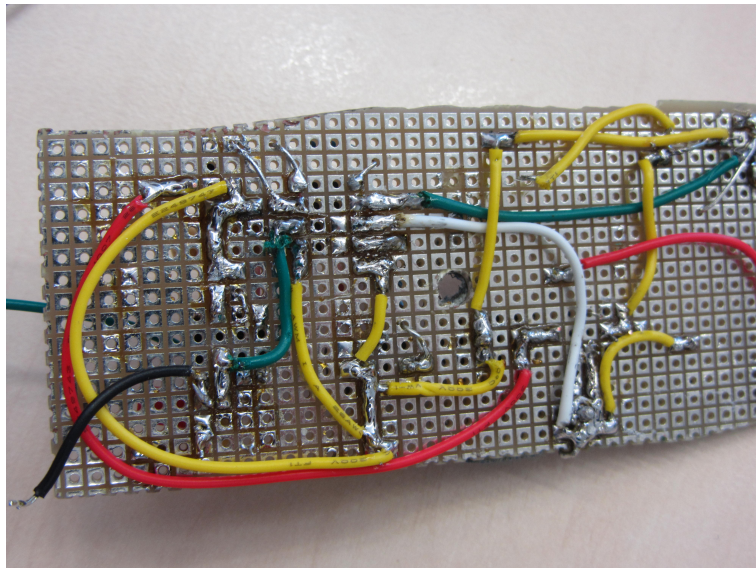


Figure A.12.: Bottom view on to the build prototype.

A. Appendix

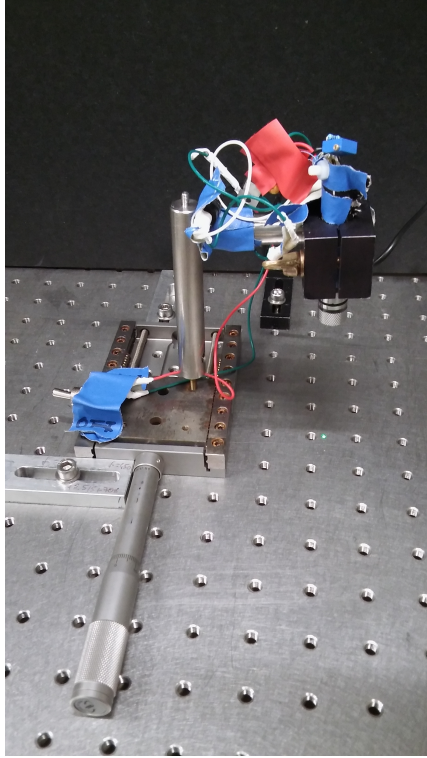


Figure A.13.: Picture of the used laser setup.

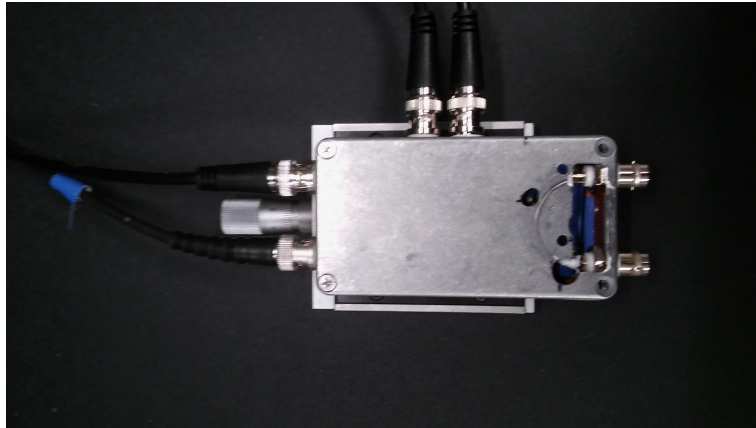


Figure A.14.: Picture of the used Detector. One can see the BNC connectors and the opening of the metal box.

A. Appendix

A.4. Data Sheets

In this section one can find the links of the data sheets for the referenced parts used.

Name	Link
Ina128P	http://www.ti.com/lit/ds/symlink/ina128.pdf
Opa37	http://www.ti.com/lit/ds/symlink/opa27.pdf
AD 842 BRZ	https://www.mouser.de/datasheet/2/609/AD8428-877830.pdf
OP 1177 ARMZ-R7	https://www.mouser.de/datasheet/2/609/OP1177_2177_4177-879575.pdf
OPA 2188 AID	http://www.ti.com/lit/ds/symlink/opa2188.pdf

Table A.5.: Name of the used parts and link to their data sheet

A.5. Further Details

The error for the different gains were calculated with Gaussian error propagation and the results are given by

$$\Delta G_1 = \Delta G_2 = \sqrt{\left(\frac{\Delta R_1}{R_2}\right)^2 + \left(\frac{\Delta R_1 R_2}{R_2^2}\right)^2}, \quad (\text{A.1})$$

$$\Delta G_{\text{total}} = \sqrt{(G_2 \Delta G_1)^2 + (G_1 \Delta G_2)^2}. \quad (\text{A.2})$$

The signal U_{before} and its error ΔU_{before} before the amplification steps are calculated by

$$U_{\text{before}} = \frac{U_{\text{sig}}}{G_{\text{total}}},$$

$$\Delta U_{\text{before}} = \sqrt{\left(\frac{\Delta U_{\text{sig}}}{G_{\text{total}}}\right)^2 + \left(\frac{\Delta G_{\text{total}} U_{\text{sig}}}{(G_{\text{total}})^2}\right)^2}, \quad (\text{A.3})$$

where U_{sig} is the signal at the oscilloscope.

The worst case for the offset U_{wc} in the used setup can be calculated by

$$U_{\text{wc}} = G_{\text{total}} U_{\text{off1}} + G_2 U_{\text{off2}}, \quad (\text{A.4})$$

where U_{off1} and U_{off2} are the maximum offset voltages of the amplifier one and two.

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“Two-Dimensional Multiwire Monitor for the Investigation of the atomic Hydrogen Beam”.
preprint.

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D. German Title, Motivation and Summary and Outlook

Anbei ist der deutsche Titel der Arbeit, sowie eine deutsche Einleitung und Zusammenfassung mit Ausblick.

D.1. Detektor für atomaren Wasserstoff

Titel der Arbeit lautet in deutsch "Detektor für atomaren Wasserstoff".

D.1.1. Motivation

Das Ziel dieser Arbeit war es, einen Detektor zu entwickeln, der es nicht nur ermöglicht, die Präsenz von atomarem Wasserstoff festzustellen, sondern auch bestimmte Eigenschaften, wie das Strahlprofil oder den Atomfluss eines solchen Wasserstoffstrahls messen kann. Dieser Detektor wurde im Rahmen des Triton-Radius Experiment, kurz T-REX, benötigt, welches das Vorhaben verfolgt, Aussagen der Quantenelektrodynamik in gebundenen Zuständen zu testen, und darüber hinaus zielt es darauf ab, Eigenschaften des Tritiumkerns zu untersuchen. [1]

Jedoch ist die Nutzung des entwickelten Detektors nicht auf eine Nutzung innerhalb von T-REX beschränkt, da sein Design eine allgemeine Verwendung in Bereichen der Physik zulässt, in denen Atomstrahlen bestehend aus Wasserstoff, Deuterium oder Tritium benutzt werden.

Die zugrundeliegende Idee von T-REX ist es, einen Strahl bestehend aus kalten Tritiumatomen zu erzeugen. Mit Hilfe eines magnetischen Multipols wird dann der langsame Anteil der Atome für das weitere Experiment selektiert und gleichzeitig zu diesem geführt. Im weiteren Verlauf treffen die Tritiumatome auf eine Wolke aus viel kälteren Lithiumatomen, welche in einer magneto-optischen Falle (MOT) gefangen sind und durch Kollisionen mit den Tritiumatomen diese weiter abbremsen sollen. So ist das abschließende Ziel, die langsamen Tritiumatome in einer speziell für sie angefertigten magnetischen Falle zu fangen und dort hochauflösende Laserspektroskopie an diesen durchzuführen. Dieses Experiment hat die Absicht, die aktuelle Messgenauigkeit für den Triton Ladungsradius um einen Faktor von 300 zu verbessern, und bietet weiterhin die Möglichkeit Kernkräfte zwischen den drei Nukleonen des Tritiumkerns zu studieren.

Da man jedoch für die ersten Tests des experimentellen Aufbaus nicht das radioaktive Element Tritium verwenden will, wird atomarer Wasserstoff verwendet. Dies kann

D. German Title, Motivation and Summary and Outlook

ohne größere Einschränkung getan werden, denn bei Tritium handelt es sich um ein Wasserstoffisotop, welches sich aus diesem Grund ähnlich verhält.[4]

Um nun das Verhalten und die Funktionsweise der einzelnen Komponenten des Experiments, wie zum Beispiel des Wasserstoffstrahls oder des magnetischen Multipols analysieren zu können, wird der in dieser Arbeit vorgestellte und getestete Detektor benötigt. Dieser soll aus diesem Grund in der Lage sein, den Wasserstoffstrahl zu analysieren. So soll er zum Beispiel ein Strahlprofil erstellen oder den Atomfluss bestimmen können, doch vor allem muss er in der Lage sein, die Anwesenheit von atomaren Wasserstoff nachweisen zu können. Also ist es das Ziel, diesen Detektor dazu zu nutzen, den experimentellen Aufbau von T-REX zu untersuchen und zu verbessern. Die aktuell verbreiteten Methoden zur Analyse eines Wasserstoffstrahls sind das Quadrupolmassenspektrometer (QMS) und die Kompressionsrohrmethode [17]. Die zuletzt genannte Methode benutzt eine Kammer, in der man den Druck mit Hilfe des Kompressionsrohrs misst, und aus den Eigenschaften des Rohrs und dem gemessenen Druck können dann die gewünschten Strahlparameter errechnet werden. Jedoch haben diese Methoden einige Nachteile. So zerstören sie den Wasserstoffstrahl während der Messung, sodass dieser nicht für weitere Experimente genutzt werden kann. Desweiteren benötigt das QMS ein hohes Vakuum, was zu Problemen mit anderen Experimenten innerhalb der Vakuumkammer führen kann. Der benötigte Platz für diese Methoden stellt zudem auch ein Problem dar, da der Platz in der Kammer beschränkt ist.

Daher ist es das Ziel, die oben erwähnten Problem mit Hilfe des Detektors zu lösen. Um dies bewerkstelligen zu können, ist die Idee, einen Detektor nach der Idee von Vassiliev et al. [19] zu konstruieren. Die Idee besteht darin, dass man die Energie, die Wasserstoff bei der Rekombination auf einer Metalloberfläche abgibt, nutzt, um einen dünnen Draht zu heizen. Dieses Heizen führt dann dazu, dass sich der Widerstand des Drahtes ändert, was dann mit einer Wheatstoneschen Messbrücke in eine messbare Spannung umgewandelt werden kann.

Aus diesem Grund wird in dieser Arbeit die notwendige Theorie des zugrunde liegenden Experiments (T-REX), sowie die benötigte Theorie, um das Funktionsprinzip des Detektors darstellen zu können, skizziert. Desweiteren werden im Theorieteil die Grundlagen der benutzten elektronischen Bauteile dargelegt. Nachdem darauf eingegangen wurde, werden die Ergebnisse der ersten mit einem Laser durchgeführten Messungen gezeigt und analysiert. Außerdem werden die mit dem Detektor gemachten Strahlprofile untersucht. Schließlich werden dann die in dieser Arbeit erzielten Ergebnisse benutzt, um eine verbesserte Version des Detektors zu entwickeln, welche auch vorgestellt wird.

Somit ist das Gesamtziel dieser Arbeit, eine Anleitung für zukünftige Projekte zu entwickeln, die den Bau und Test eines einfachen und kostenarmen Detektors für Strahlen bestehend aus atomaren Wasserstoff, Tritium oder Deuterium vorstellt.

D.1.2. Zusammenfassung und Ausblick

Zunächst legt die Analyse des benutzten Lasers nahe, dass es für zukünftige Experimente ratsam wäre, einen Laseraufbau zu benutzen, der einen wohlgeformteren Strahl erzeugt und den man besser in seiner Leistung einstellen kann, damit vergleichbarere Resultate erzeugt werden können, und um Unsicherheiten in der Analyse zu vermeiden. Dennoch konnte der Laser benutzt werden, um den Wasserstoffstrahl zu simulieren, da das fundamentale Prinzip, welches darin besteht, den Messdraht zu erwärmen, auch durch den Laser gegeben ist. Da weiterhin festgestellt wurde, dass der Laser in seiner Leistung nicht all zu stark von der zu erwartenden Leistung des Wasserstoffstrahls abweicht, können die in dieser Arbeit erzielten Ergebnisse als ersten Beweis angesehen werden, dass die Funktionsweise des vorgeschlagenen Detektors funktioniert und die gewünschten Ziele erreicht werden können.

Weiterhin hat die Analyse des Detektors gezeigt, dass Spannungen im nV-Bereich registriert werden können, und dass es möglich ist, die gewünschte Sensitivität zu erreichen. Desweiteren konnte eine Übereinstimmung zwischen angewandter Theorie und der Messdaten gefunden werden. Dies führt dazu, dass man die Aussagen der Theorie dazu nutzen kann Vorhersagen zu berechnen, die bei der Entwicklung verbesserter Designs zur Hilfe genommen werden können. Außerdem war es im Rahmen dieser Arbeit möglich zu zeigen, dass der Detektor dazu benutzt werden kann Strahlprofile zu erstellen, aus denen man Charakteristiken des Strahls berechnen kann, die mittels anderer Methoden verifiziert werden konnten. Die Analyse lässt jedoch vermuten, dass die aktuelle Version des Detektors nicht fähig ist, in ausreichender Genauigkeit den Wasserstoffstrahl zu untersuchen. Aus diesem Grund wurden die in dieser Arbeit festgestellten Probleme am aktuellen Design dazu benutzt, eine verbesserte Version zu erstellen. So wurde in dieser die Messbrücke symmetrischer gestaltet und in ihrer Spannungsversorgung stabilisiert, um Fluktuationen im Signal zu reduzieren. Desweiteren wurde die absolute Verstärkung des Detektors erhöht. Außerdem wurde, um eine Sättigung in den genutzten Verstärkern zu vermeiden, eine Technik implementiert, die das Ausgleichen der Offsetspannung der Verstärker zulässt. Schließlich ist es geplant, den Untergrund unter dem Messdraht auf der Detektorplatine zu entfernen, um zu verhindern, dass es zu einer Strahlreflektion durch diese kommt.

So ist es geplant, die verbesserte Version zu bauen und mit dieser Tests, ähnlich zu denen in dieser Arbeit, durchzuführen. Darüber hinaus sollten weitere Experimente, wie zum Beispiel der Test mit einem Funktionsgenerator, durchgeführt werden, um weitere Eigenschaften des Detektor zu überprüfen. So sollte die absolute Verstärkung getestet werden oder auch das Detektorverhalten bei verschiedenen Temperaturen. Aber das abschließende Ziel ist es, die verbesserte Version mit dem Wasserstoffstrahl zu testen, um abschließend zu klären, ob der Detektor die für ihn vorgesehenen Aufgaben erfüllen kann. So sollte zudem nach den ersten Wasserstoff-Tests ein Bewegungssystem entwickelt werden, welches es ermöglicht, den Detektor in der Vakuumkammer zu bewegen, um so Strahlprofile aufzunehmen.